Latent Spectral Dynamics: Uncovering Truthfulness in Large Language Models via Koopman Operator Theory

Anonymous authors
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Abstract

Autoregressive large language models (LLMs) have been the subject of extensive interpretability research through static representations. Yet a dynamical-systems account of their global, token-wise activation trajectories remains underdeveloped, leaving the holistic temporal evolution of these models largely uncharacterized. Leveraging Koopman operator theory, we formulate token-wise residual stream states as a stochastic, autonomous, discrete-time dynamical system. We utilize time-delay embeddings to reconstruct a Markovian state, and approximate its Koopman operator via Hankel-Extended Dynamic Mode Decomposition (Hankel-EDMD). On generation trajectories from models of different sizes under truthful and untruthful prompting regimes, we find that Koopman operator approximations using coordinate bases yield accurate short-horizon predictions and stable multi-step rollouts, suggesting Markovian transitions within the constructed states. General analyses show that bulk covariance and effective rank remain comparable across both regimes, indicating a shared global latent backbone. In stark contrast, the learned Koopman operators exhibit markedly distinct dynamical modes. Truthful trajectories converge toward a stable invariant measure dominated by near-unity real eigenvalues. Untruthful generation, however, systematically escapes this stable basin, driven by oscillatory, complex-conjugate eigenvalues. Cross-domain validation on unlabeled questions confirms that this oscillatory signature tracks epistemic states rather than the truthfulness labels, generalizing without supervision. By grounding these findings in Koopman spectral analysis, these results establish Latent Spectral Dynamics as a principled framework for investigating LLM internals.

1 Introduction

Autoregressive large language models (LLMs) are increasingly entrusted with high-stakes decisions, yet their internal processes remain opaque, creating uncertainty that directly manifests in a growing catalogue of failure modes. Models *hallucinate*, generating fluent but factually ungrounded text that is hard to distinguish from truthful output (Huang et al., 2024; Tonmoy et al., 2024). They *sycophantically deceive*, tailoring responses to perceived user expectations rather than reporting internal assessments (Greenblatt et al., 2024; Hubinger et al., 2024). And they succumb to adversarial *jailbreaks*, bypassing safety guardrails under carefully crafted prompts (Zou et al., 2023b; Qi et al., 2024). Pathologies as such have predominantly been studied as independent phenomena, each addressed by specialized benchmarks and mitigation strategies. We hypothesize, however, that these failure modes share a common mathematical foundation within the model's activations: viewed through a dynamical-systems lens, they can be understood as features of the underlying geometry, characterized by the spectral signatures of the transition operator.

This perspective motivates our central question: *can LLM alignment failures be diagnosed as dynamical instabilities in latent activation space?* While extending this framework across the full taxonomy of such failures remains an open challenge, here we focus on a fundamental instantiation: the trajectory-level divergence between truthful and hallucinated generation. Through this lens, we find strong empirical evidence that hallucination emerges as a

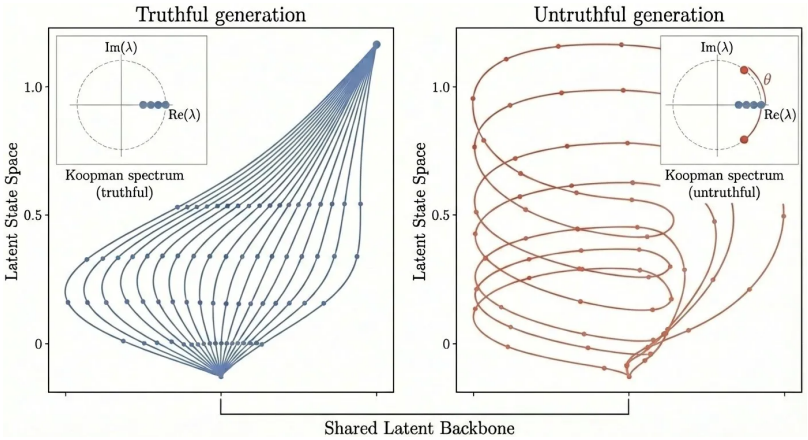

Figure 1: Both truthful (blue) and untruthful (red) generation trajectories originate from a shared latent backbone in the residual stream. Truthful trajectories converge toward a stable invariant measure, governed by near-unity real Koopman eigenvalues. Untruthful trajectories exhibit structured oscillatory dynamics driven by complex-conjugate eigenvalue pairs. Dots mark discrete token steps. This divergence is undetectable from static activation snapshots and emerges only through temporal embedding of the residual stream.

trajectory-level divergence, in which the model's internal state exits a stable generative basin and exhibits oscillatory dynamics over token-by-token generation —a dynamical signature that, as we show, generalizes beyond the labeled setting to domains without ground truth.

Mechanistic interpretability has produced powerful tools for reverse-engineering Transformer internals—including circuit analysis (Elhage et al., 2021; Naseem, 2026), probing methods (Belrose et al., 2023; Ghandeharioun et al., 2024), representation engineering (Park et al., 2023; Zou et al., 2023a), and sparse autoencoders (Wang et al., 2024)—yet these approaches remain *static*: they analyze fixed layers, token positions, or weight states rather than token-to-token computation. Although a small but growing body of work has adopted dynamical and spectral perspectives on neural computation (Bai et al., 2019; Chen et al., 2018; Xie et al., 2021; Zekri et al., 2024; Geng et al., 2026; Jha & Reagen, 2026; Li et al., 2025a; Noël, 2026), no prior work has systematically extracted the transition operator governing the temporal evolution of an LLM's latent space during autoregressive generation.

We fill this gap by formulating token-wise LLM activations as a stochastic, autonomous, discrete-time dynamical system and analyzing it through the lens of Koopman operator theory. By constructing a time-delay embedding of final-layer residual stream vectors, we reconstruct a fixed-dimensional latent state that approximately preserves Markovian dynamics. The system's nonlinear state transitions are then lifted into an infinite-dimensional linear space of observables governed by the Koopman operator. Finally, we obtain a finite-dimensional approximation of this operator using Hankel-Extended Dynamic Mode Decomposition (Hankel-EDMD) with Tikhonov-regularized dual ridge regression.

Analyzing generation trajectories from Qwen2.5-14B-Instruct under truthful and untruthful prompting regimes—with cross-domain validation on unlabeled philosophical questions—we find markedly distinct spectral signatures. Our central contributions are:

1. **Generalizable Dynamical Systems Framework**

   We formulate autoregressive LLM generation as a stochastic dynamical system on time-delay-embedded residual-stream states, and find that this construction supports Markovian transitions: coordinate-basis Koopman operators yield stable autonomous rollouts across all tested conditions. This provides the first validation of a dynamical model of the residual stream in continuous activation space, establishing a mathematically grounded framework in which distinct generative behaviors correspond to distinct spectral signatures of the transition operator.

2. **Spectral Signatures of Truthful and Untruthful Generation.**
   We demonstrate that truthful and untruthful generation trajectories yield distinct Koopman spectra (Figure 1) under the same model. Truthful generation converges toward an invariant measure with near-unity real eigenvalues. Untruthful generation exhibits oscillatory dynamics with eigenvalues tracing distinct arcs in the complex plane. Cross-domain validation on unlabeled philosophical questions suggests that the mode tracks the model's epistemic state rather than the truthfulness label, extending beyond the labeled truthful-untruthful setting without supervision.

3. **Static Indistinguishability, Dynamical Divergence.**
   Truthful and untruthful trajectories are statistically indistinguishable under standard representational diagnostics – matched covariance structure, overlapping effective rank, shared dominant PCA modes – yet yield different Koopman operators. The oscillatory signature emerges only through temporal embedding, establishing that the divergence is dynamical and inaccessible to static methods.

2 Related Work

Spectral and Topological Analysis of Transformer Internals. Circuit analysis (Elhage et al., 2021; Naseem, 2026), probing classifiers (Belrose et al., 2023), and representation engineering (Zou et al., 2023a) have produced foundational insights into Transformer internals but still treat the model as static. Recent spectral approaches move toward dynamical characterization: Geometry of Reason (Noël, 2026) analyzes graph-Laplacian structure in attention, while NerVE (Jha & Reagen, 2026) tracks feed-forward eigenspectrum dynamics across depth. Yet both remain tied to static architectural snapshots—frozen weights or per-layer graph topologies—rather than the temporal trajectories of autoregressive generation.

Frequency-Domain and Embedding-Level Hallucination Detection. HSAD (Li et al., 2025a) treats hidden-layer vectors sampled across Transformer *layers* as a discrete-time signal and applies FFT to isolate hallucination-correlated spectral anomalies. Akrout (2023) apply DMD to sentence-level embeddings, finding that hallucinated paragraphs exhibit a persistently low-rank spectrum. While both provide evidence that hallucination manifests as spectral instability, HSAD operates across depth within a single forward pass (discarding phase information and transition operators), and Akrout's analysis uses sentence embeddings rather than internal activations without reconstructing the underlying dynamical system. Our Koopman framework recovers the full spectral decomposition—eigenvalues, eigenfunctions, and modes—of the nonlinear transition map that *generates* those frequencies.

Dynamical Systems Perspectives on Neural Computation. Deep Equilibrium Models (Bai et al., 2019) and Neural ODEs (Chen et al., 2018) established that neural computation can be analyzed as a dynamical system. More recently, in-context learning has been framed as implicit Bayesian inference (Xie et al., 2021), implying that sufficiently directive prompts can render generation locally Markovian—a property used in finite-state Markov-chain models of bounded-context LLMs (Zekri et al., 2024; Geng et al., 2026). However, these models operate over the discrete output space and discard the activation trajectories analyzed here.

Koopman Operator Theory and Applications to Sequence Models. Koopman operator theory (Koopman, 1931; Mezić, 2005) lifts nonlinear dynamics into a linear observable space, with practical approximations via DMD and extensions (Schmid, 2010; Williams et al., 2015). Koopman-inspired methods have recently emerged in machine learning. ReLaX (Zhang et al., 2025a) linearizes hidden-state dynamics via a neural Koopman dictionary during RLVR-based policy optimization, introducing Dynamic Spectral Dispersion to quantify latent heterogeneity. MamKO (Li et al., 2025b) combines Koopman modeling with Mamba-inspired operator generation for time-varying nonlinear control. KoopSTD (Zhang et al., 2025b) develops a Koopman-spectral similarity metric with explicit timescale decoupling for dynamical systems. While these works apply Koopman methods to control, optimization, similarity, or training-time diagnostics, ours is the first to estimate a Koopman operator directly from an LLM's autoregressive trajectory to diagnose *inference-time* alignment failures.

3 Methodology

This framework recasts autoregressive LLM generation as a *stochastic autonomous discrete-time dynamical system*. This provides a mathematically grounded approach to diagnosing behavioral divergence through spectral signatures of the approximated Koopman operator.

3.1 Markovian Dynamics of Autoregressive Generation

Standard analyses treat autoregressive generation as a highly non-Markovian process, factorizing the joint probability of N tokens as $P_\phi(x_{1:N}) = \prod_{k=1}^{N} P_\phi(x_k \mid x_{<k})$. To formulate an autonomous dynamical system, we require a *Markovian state*—one that renders the future conditionally independent of the past. In fact, the Transformer architecture provides such a state through the internal Key-Value (KV) cache.

At generation step k, the model maintains a KV cache $\mathcal{S}_k = \{(K_\ell^{(j)}, V_\ell^{(j)})\}_{\ell,j}$ containing key-value pairs across all layers ℓ and positions $j \leq p+k$. A deterministic forward pass produces the final-layer residual stream vector $h_k = f_\phi^{(L)}(\mathcal{S}_k) \in \mathbb{R}^d$, from which the logit vector is computed via a linear unembedding: $l(h_k) = W_U h_k + b_U \in \mathbb{R}^{|\mathcal{V}|}$. The sampled token $x_{k+1} \sim \mathrm{Softmax}(l(h_k)/\tau)$ then triggers a deterministic cache update $\mathcal{S}_{k+1} = U_\phi(\mathcal{S}_k, x_{k+1})$.

Note that the next-token distribution depends on the full history $x_{1:k}$ only through $\mathcal{S}_k$:

$$P_\phi(x_{k+1} \mid x_{1:k}) \;=\; P_\phi(x_{k+1} \mid \mathcal{S}_k). \tag{1}$$

Since $\mathcal{S}_{k+1} = U_\phi(\mathcal{S}_k, x_{k+1})$ is deterministic given $(\mathcal{S}_k, x_{k+1})$, and x_{k+1} is drawn from a distribution depending only on $\mathcal{S}_k$, the next cache state $\mathcal{S}_{k+1}$ is conditionally independent of all prior states given $\mathcal{S}_k$. The sequence $\{\mathcal{S}_k\}$ therefore constitutes an *exact Markov chain*(Appendix A.1).

3.2 Phase Space Reconstruction via Delay Embedding

Although $\mathcal{S}_k$ is an exact Markov state, it is impractically high-dimensional for operator estimation. We construct a fixed-dimensional surrogate by delay-embedding the residual stream $h_k \in \mathbb{R}^d$ into a Hankel state of window size W:

$$\mathbf{H}_k = \left[h_{k-W+1}^\top, \;\ldots,\; h_k^\top\right]^\top \in \mathbb{R}^{Wd} \tag{2}$$

Our goal is to establish that $\mathbf{H}_k$ approximates $\mathcal{S}_k$ as a predictive state:

$$P_\phi(x_{k+1} \mid x_{1:k}) \;=\; P_\phi(x_{k+1} \mid \mathcal{S}_k) \;\approx\; P_\phi(x_{k+1} \mid \mathbf{H}_k) \tag{3}$$

where the first equality is exact by (1), and the second is supplied by delay embedding.

3.2.1 *Assumption: Low-dimensional Concentration*

The exact Markov state $\mathcal{S}_k$ resides in $\mathbb{R}^{D_{\max}}$, where $D_{\max} = (p + L_{\max}) \cdot 2L \cdot n_{\text{heads}} \cdot d_{\text{head}}$. Operator estimation in this ambient space is intractable; to make the delay-embedding strategy of Eq. (2) viable, we posit that the dynamics are governed by far fewer degrees of freedom than the ambient dimension suggests.

Assumption 1 *Under a fixed prompting regime θ^*, generation trajectories concentrate on a submanifold $\mathcal{M}_{\theta^*} \subset \mathcal{M}$ of intrinsic dimension $m_{\text{eff}} \ll D_{\max}$.*

This is standard for phase-space reconstruction: the required embedding dimension scales with m_{eff} instead of $D_{\max}$.

3.2.2 *Time-delay Embedding Guarantee*

For the delay-embedded state $\mathbf{H}_k$ to support spectral analysis, the mapping from cache trajectories to Hankel states must preserve the statistical structure of the underlying system. We define the *delay-coordinate map* $\Phi_W : \mathcal{M}^W \to \mathbb{R}^{Wd}$, which applies the observation map $g = f_\phi^{(L)}$ to each element of a length-W trajectory segment and concatenates the results:

$$\Phi_W(\mathcal{S}_{k-W+1}, \ldots, \mathcal{S}_k) = \left[g(\mathcal{S}_{k-W+1})^\top, \ldots, g(\mathcal{S}_k)^\top\right]^\top = \mathbf{H}_k. \tag{4}$$

The key requirement is that the *pushforward* $(\Phi_W)_\# \mu$—the distribution over Hankel states induced by the invariant measure μ of the cache Markov chain (§3.1)—uniquely determines μ. This is guaranteed by the measure-theoretic phase-space reconstruction theorem of Botvinick-Greenhouse et al. (2025): for a compact sub-manifold $\mathcal{M}_{\theta^*}$ of intrinsic dimension m_{eff}, a Lipschitz observation map g, and a window size $W \geq 2\, m_{\text{eff}} + 1$, the pushforward $(\Phi_W)_\# \mu$ uniquely determines μ for generic g.

This uniqueness places $L^2(\mu)$ on $\mathcal{M}_{\theta^*}$ in correspondence with $L^2((\Phi_W)_\# \mu)$ on $\mathbb{R}^{Wd}$, so any linear operator governing the evolution of observables can be estimated equivalently from $\{\mathbf{H}_k\}$. Since the next-token distribution $P_\phi(x_{k+1} = v \mid \mathcal{S}_k)$ lies in $L^2(\mu)$, this establishes Eq. (3). Compactness of $\mathcal{M}_{\theta^*}$ and Lipschitz regularity of $g = f_\phi^{(L)}$ are verified in Appendices A.2 and A.3, which also discuss the theorem's scope and the gap between its formal window-size bound and our practical choices.

3.2.3 *Approximation through Practical Window Sizes*

Our experimental windows ($W \in \{1, 3, 5, 8\}$) likely fall below the formal bound $2\, m_{\text{eff}} + 1$ due to computational constraints. To bridge the gap between formal guarantees and practical performance, we introduce the following empirical criterion.

Definition 1 (ε-Predictive Embedding) Φ_W *is an* ε-predictive embedding *if the estimated operator* $\mathbf{K}$ *achieves held-out relative one-step MSE* $\leq \varepsilon$*:*

$$\text{Rel-MSE}(1) \;:=\; \frac{\sum_i \|\mathbf{H}_{k_i+1} - \mathbf{H}_{k_i}\mathbf{K}\|_2^2}{\sum_i \|\mathbf{H}_{k_i+1}\|_2^2} \;\leq\; \varepsilon.$$

An ε*-predictive embedding captures at least fraction* $(1-\varepsilon)$ *of the predictable transition variance, ensuring the dominant eigenvalues are well-estimated; in Table 1, the coordinate-basis results* ($\varepsilon \approx 0.17$–0.31*, stable rollouts through* h=8*) satisfy this criterion.*

Sections 3.2.1–3.2.3 together establish that $\mathbf{H}_k$ is an approximate predictive state for $\{\mathcal{S}_k\}$:

1. **State Representation:** $P_\phi(x_{k+1} \mid x_{1:k}) = P_\phi(x_{k+1} \mid \mathcal{S}_k) \approx P_\phi(x_{k+1} \mid \mathbf{H}_k)$, guaranteed by the phase-space reconstruction theorem for $W \geq 2m_{\text{eff}}+1$ and supported empirically via Definition 1 for practical W.
2. **Markovian Transition:** $\{\mathbf{H}_k\}$ inherits approximate Markovian dynamics from $\{\mathcal{S}_k\}$, with approximation quality bounded by the ε-predictive criterion.

3.3 Autonomous Dynamics Formulation

Autoregressive generation alternates between continuous and discrete domains: the continuous state $\mathbf{H}_k$ parameterizes the emission of a discrete token x_{k+1}, which feeds back to produce the next continuous state $\mathbf{H}_{k+1}$. To obtain a self-contained dynamical system on a continuous state space, we must eliminate the intermediate discrete variable.

Under the approximate state representation established in (3), the next-token probability is the temperature-scaled softmax applied to the latest residual stream component h_k of $\mathbf{H}_k$:

$$P_\phi(x_{k+1} = v \mid \mathbf{H}_k) \;=\; \text{Softmax}_v\left(\frac{l(h_k)}{\tau}\right), \qquad l(h_k) = W_U\, h_k + b_U \in \mathbb{R}^{|\mathcal{V}|} \tag{5}$$

We apply the *Gumbel-Max reparameterization* to decouple the categorical stochasticity into an additive exogenous noise vector $\Xi_k \sim \text{Gumbel}(0, \mathbf{I}_{|\mathcal{V}|})$, expressing the sampled token as a deterministic function of continuous inputs:

$$x_{k+1} = \arg\max_{v \in \mathcal{V}}\left(\frac{l(h_k)_v}{\tau} + \Xi_{k,v}\right) \;\equiv\; \mathcal{T}_{\tau,\phi}(\mathbf{H}_k, \Xi_k) \tag{6}$$

Upon generating x_{k+1}, the Transformer computes h_{k+1} via a deterministic forward pass on the updated cache. Substituting $\mathcal{T}_{\tau,\phi}$ into this update and shifting the Hankel window

eliminates the discrete token entirely, finally yielding a **stochastic autonomous discrete-time dynamical system** (derivation in Appendix A.4):

$$\mathbf{H}_{k+1} = \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \Xi_k) \tag{7}$$

Under the delay embedding of §3.2, $\mathbf{H}_k$ approximately determines $\mathcal{S}_k$, so $\mathbf{H}_{k+1}$ depends on $\mathbf{H}_k$ and the exogenous noise Ξ_k alone. The sequence $\{\mathbf{H}_k\}$ therefore constitutes an approximate Markov chain on $\mathbb{R}^{Wd}$, governed by the transition kernel:

$$\mathcal{P}_{\theta^*}(d\mathbf{H}_{k+1} \mid \mathbf{H}_k) = \mathbb{P}_{\Xi}\big(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \Xi_k) \in d\mathbf{H}_{k+1}\big) \tag{8}$$

The transition map $\mathcal{F}_{\phi,\theta^*}$ and its kernel $\mathcal{P}_{\theta^*}$ are indexed by the prompting regime θ^*, reflecting the empirical observation that distinct regimes (e.g., factual vs. fabricated generation) yield distinct operators.

3.4 The Koopman Operator and Finite Approximation

To analyze the Markovian transition map, we lift the nonlinear dynamics into an infinite-dimensional Hilbert space of observables $L^2(\mu)$ using the Koopman operator $\mathcal{K}_{\theta^*}$. This linear operator dictates the temporal evolution of an observable function g via conditional expectation over the Markov kernel:

$$(\mathcal{K}_{\theta^*} g)(\mathbf{H}_k) = \mathbb{E}_{\mathbf{H}_{k+1} \sim \mathcal{P}_{\theta^*}(\cdot \mid \mathbf{H}_k)}[g(\mathbf{H}_{k+1})] = \mathbb{E}_{\Xi_k}[g(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \Xi_k))] \tag{9}$$

Notably, as the sampling temperature $\tau \to 0^+$, the stochastic forcing Ξ_k collapses. The system becomes purely deterministic ($\mathbf{H}_{k+1} = \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k)$), reducing the Markov kernel to a Dirac delta measure and the operator to exactly $(\mathcal{K}_{\theta^*} g)(\mathbf{H}_k) = g(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k))$. Any finite-dimensional approximation projects $\mathcal{K}_{\theta^*}$ onto the span of a chosen observable dictionary, recovering the component of the dynamics visible through those observables.

We empirically approximate this operator using **Hankel-Extended Dynamic Mode Decomposition** (Hankel-EDMD) with three observable dictionaries (§4). Given N transition pairs, we construct data matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{N \times D}$ (where D is the observable dimension):

$$\mathbf{X} = \begin{bmatrix} g(\mathbf{H}_0)^\top \\ \vdots \\ g(\mathbf{H}_{N-1})^\top \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} g(\mathbf{H}_1)^\top \\ \vdots \\ g(\mathbf{H}_N)^\top \end{bmatrix},$$

where $g(\mathbf{H}_k) = [g_1(\mathbf{H}_k), \ldots, g_D(\mathbf{H}_k)]^\top \in \mathbb{R}^D$. For the coordinate basis used in our main results, $g(\mathbf{H}_k) = \mathbf{H}_k$ and $D = Wd$. The Koopman matrix $\mathbf{K}_{\text{primal}} \in \mathbb{R}^{D \times D}$ satisfies $\mathbf{Y} \approx \mathbf{X}\mathbf{K}_{\text{primal}}$ and is obtained by minimizing a Tikhonov-regularized least-squares objective. Because $D \gg N$ in the coordinate basis, we employ a **Dual Ridge formulation** that factorizes the problem within the $N \times N$ sample space, entirely bypassing the high-dimensional D (full derivation in Appendix A.5).

4 Experiments

4.1 Datasets and Trajectory Generation

To validate the dynamical framework and test whether the discovered spectra are robust to syntax and domain, we construct trajectories in two generative regimes. **(1) Primary labeled dataset :** We extract final-layer residual streams ($d = 5120$) from Qwen2.5-14B-Instruct (Yang et al., 2024) using $P = 200$ paired prompts eliciting factual versus fabricated continuations (see Appendix C, $L_p = 32$ tokens, $\tau \in \{0.0, 0.3, 0.7\}$). **(2) Cross-domain validation (ab20):** To test whether the modes track epistemic uncertainty rather than supervised truthfulness labels, we generate ∼900 responses to open philosophical questions (see appendix D, $\tau \in \{0.3, 1.0\}$).

Label	Basis	Train MSE	Held-out 1-Step	Rollout h=1	Rollout h=5	Rollout h=8
Untruthful	PCA	3.06	6.74	205.30	1874.44	2061.67
	Coordinate	**0.04**	**1.50**	**1.43**	**4.07**	**5.03**
	Fourier	1440.89	1827.97	27279.50	76551.09	65739.39
Truthful	PCA	3.32	9.38	178.66	616.17	627.94
	Coordinate	**0.07**	**2.28**	**2.16**	**5.19**	**6.07**
	Fourier	1645.80	2509.23	28835.56	37845.39	31901.91

Table 1: Model fit and multi-step predictability (W=5, τ=0, P=100, α=10000). The coordinate basis achieves rollout errors two orders of magnitude below PCA and four below Fourier, remaining stable through h=8. Multi-step rollout curves appear in Appendix E.

4.2 State Space Reconstruction and Observable Dictionaries

For each trajectory, we reconstruct an approximately Markovian state with a delay-embedding window $W \in \{1, 3, 5, 8\}$, initializing $\mathbf{H}_0$ with W prepended prompt tokens. This yields N_{train} valid transition pairs $(\mathbf{H}_k, \mathbf{H}_{k+1})$ per class. Before Hankel concatenation, raw activations h_k are transformed by an observation map $g(h_k)$. We study three observable dictionaries: *(i) Coordinate basis* ($g(h_k) = h_k$): uncompressed dynamics in the dual Koopman regime ($D \gg N$). *(ii) PCA basis* ($g(h_k) \in \mathbb{R}^m$): compressed dynamics in the primal regime ($m = 256$ for baselines, $m = 64$ for scaling experiments). *(iii) Fourier basis* ($g(h_k) \in \mathbb{C}^c$): the first $c = 32$ low-frequency DFT coefficients (Appendix B).

4.3 Operator Estimation and Rollout Fidelity

We estimate class-conditional Koopman operators ($\mathbf{K}_{\text{truth}}$ and $\mathbf{K}_{\text{lie}}$) via Hankel-EDMD. For the overparameterized coordinate basis, Tikhonov regularization α is tuned by grid search ($\alpha \in [10^3, 10^4]$). To test the Markovian approximation, we evaluate autonomous multi-step rollouts: given $\mathbf{K}$ and test state $\mathbf{H}_k$, we iterate $\widehat{\mathbf{H}}_{k+n}^\top = \mathbf{H}_k^\top \mathbf{K}^n$ without teacher forcing. Fidelity is measured by **relative multi-step rollout MSE**: Rel-MSE$(n) = \sum_i \|\mathbf{H}_{t_i+n} - \widehat{\mathbf{H}}_{t_i+n}\|_2^2 \,/\, \sum_i \|\mathbf{H}_{t_i+n}\|_2^2$, which normalizes across observable bases of different scale.

4.4 Oscillatory Mode Detection and Robustness

Following the eigendecomposition of $\mathbf{K}$, we identify the dynamical signature of epistemic uncertainty through anomalous *oscillatory Koopman modes*: complex-conjugate eigenvalue pairs separated from the near-real, slow-decay backbone by their rotational dynamics. A mode is flagged when $|\lambda| > 0.80$, $\theta \in (0.10, 0.45)$ rad, and $|\operatorname{Im}(\lambda)| > 0.05$. To test whether this signature is structural rather than numerical, we repeat the estimation pipeline under an N-scaling protocol with $N \in \{100, 150, 200, 250\}$ training sequences and $K = 3$ seeds per N, and use Fisher's exact test to assess class-level enrichment across the resulting grid.

5 Results and Analysis

We organize the results around a central hypothesis: truthful and untruthful generations share a common coarse dynamical backbone and differ primarily in a localized oscillatory Koopman mode that emerges only under temporal embedding. We first validate the dynamical framework (§5.1), establish the shared backbone (§5.2), present the core spectral discovery (§5.3), and confirm cross-domain generalization (§5.4).

5.1 Dynamical Fit Quality and Observable Basis Selection

We compare EDMD operators fit in coordinate, PCA, and Fourier observable bases. Table 1 reports the full comparison. The coordinate basis is the only representation that is simultaneously accurate in one-step prediction and stable under autonomous rollout: it achieves relative rollout MSE two orders of magnitude below PCA and four below Fourier, remaining stable through horizon h=8. All subsequent analyses therefore use the coordinate basis. Notably, untruthful trajectories are consistently 30–43% more predictable than truthful ones under the linear Koopman model across all window sizes and rollout horizons, despite sharing a spectral radius near 0.99—providing eigenvalue-free evidence of greater linear regularity in untruthful dynamics (Appendix F).

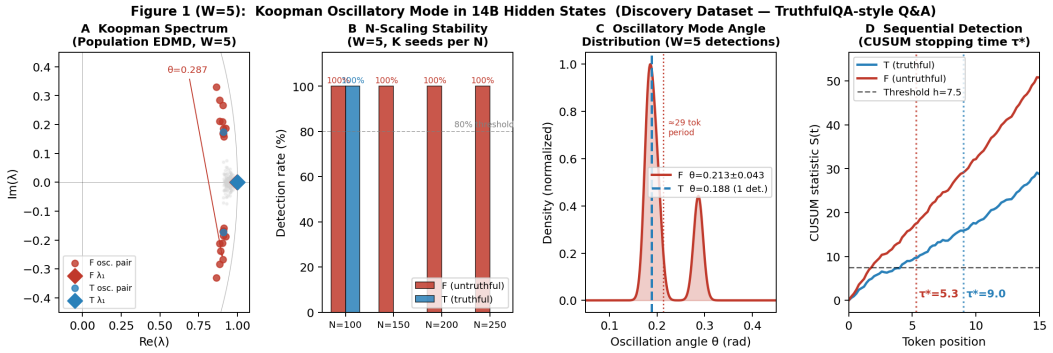

Figure 2: **Core discovery (14B). (A)** Koopman spectrum: untruthful dynamics contain a complex-conjugate pair at $\theta \approx 0.23$ rad absent from the truthful spectrum. **(B)** N-scaling stability: detection is robust across sample sizes for F but sparse for T. **(C)** Phase distribution: untruthful detections cluster at $\theta \approx 0.20$ rad; truthful detections are dispersed. **(D)** CUSUM sequential detection: untruthful trajectories enter the oscillatory regime $\sim$3.7 tokens earlier.

5.2 Shared Global Latent Backbone

We next ask whether the two regimes differ at the level of global latent geometry. The answer is largely no. Figures 6 and 7 (Appendix G) show overlapping distributions of Marchenko–Pastur spike energy fraction (0.879 ± 0.064 vs. 0.896 ± 0.073), effective rank (31.0 ± 10.6 vs. 29.0 ± 12.5), and largest-spike ratio. These statistics characterize the *global* covariance structure of the full activation space: they upper-bound the number of high-variance directions but do not directly estimate the intrinsic dimension m_{eff} (§3.2). Cross-basis dominant-mode cosine similarities support the shared-backbone picture: 0.921 (Fourier), 0.574 (PCA), and 0.193 (coordinate), indicating shared coarse structure with divergence only at full resolution. Running effective rank shows similar convergence (28.8 vs. 28.1; Appendix G). This negative result rules out the simplest interpretation that untruthful generations are "more random" or globally less structured, motivating finer spectral analysis.

5.3 The Oscillatory Koopman Mode

The class-specific signal appears only in the Koopman spectrum. Across all window sizes $W \in \{3, 5, 8\}$, the untruthful Koopman operator consistently exhibits complex-conjugate eigenvalue pairs with oscillatory phases in the range $\theta \approx 0.20$–0.23 rad (period $\approx$27–31 tokens), while the truthful operator is dominated by near-unity real eigenvalues. Fig. 2A illustrates this at W=5: the untruthful spectrum contains a cluster of complex-conjugate modes absent from the truthful spectrum, while the leading eigenvalues remain comparable—$\lambda_1 \approx 0.99$ in both classes—confirming the shared backbone.

N-Scaling Stability and Statistical Significance. We assess robustness via an N-scaling protocol (N $\in \{100, 150, 200, 250\}$, K=3 seeds per N). At W=5, the oscillatory mode is detected in 100% of untruthful runs at all sample sizes, while truthful detection remains near zero (Fig. 2B). The untruthful phase clusters at $\theta = 0.213 \pm 0.043$ rad (Fig. 2C). This enrichment is highly significant: Fisher $p < 10^{-5}$ for $W \geq 3$ across all conditions (Appendix I). Critically, the mode is not detectable at W=1 (Fisher $p = 0.17$, n.s.), establishing that the signal is genuinely temporal—consistent with the delay-embedding requirement of §3.2.

Sequential Onset. To characterize when the oscillatory mode activates during generation, we apply a CUSUM change-point detector to the squared projection energy onto the oscillatory subspace V_{lie} (defined in Appendix J). Untruthful trajectories cross the detection threshold at $\tau^* = 5.3$ tokens on average, compared to 9.0 for truthful trajectories (Fig. 2D)—a $\sim$3.7-token earlier onset consistent with untruthful generation entering the oscillatory regime sooner. Importantly, V_{lie} alone achieves only modest static classification (AUC $= 0.676$), confirming that the oscillatory subspace carries a distinct *dynamical* signature rather than simply encoding the dominant direction of class separability (Appendix J).

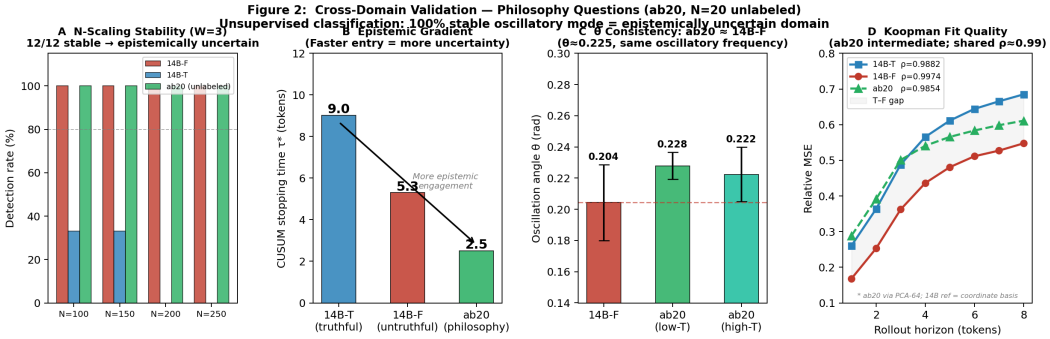

Figure 3: **Cross-domain validation (ab20). (A)** N-scaling: 12/12 detections at both temperatures. **(B)** Epistemic gradient: τ^* decreases from T $\rightarrow$ F $\rightarrow$ ab20. **(C)** θ consistency with 14B-F. **(D)** Koopman fit quality: ab20 is intermediate between 14B-T and 14B-F.

5.4 Cross-Domain Generalization (ab20)

The 14B discovery raises a natural question: does the oscillatory mode generalize beyond the *labeled T/F setting*, or is it specific to that prompting regime? We test this on ab20 philosophical questions—a domain with no ground truth—so any detected mode cannot be attributed to supervised labels.

The oscillatory pair is detected in *all* 12/12 trials at both sampling temperatures (Fig. 3A), with $\theta = 0.225 \pm 0.014$ rad, bracketing the 14B-F reference. This 100% stability exceeds 14B-F (85.7%), consistent with stronger epistemic demands in philosophical questions. The CUSUM stopping time reveals a continuous gradient (Fig. 3B): $\tau^*_T = 9.0 \rightarrow \tau^*_F = 5.3 \rightarrow \tau^*_{\text{ab20}} = 2.5$ tokens, spanning confident factual retrieval to genuine domain uncertainty. Koopman fit quality lies between the two labeled 14B conditions (Fig. 3D), placing ab20 in a distinct dynamical regime; full spectral results appear in Appendix K.

These results establish that the oscillatory mode is not a deception artifact but a genuine dynamical property of the generation process that generalizes beyond the labeled setting: it appears most strongly where no definitive answer exist. Scaling preliminary experiments on Qwen2.5-32B and 72B confirm that the mode persists with a conserved frequency ($\theta \approx 0.19$ rad across all three scales), though the class-discriminative pattern depends on the epistemic demands of the task format; a full analysis appears in Appendix L.

6 Conclusion

We have introduced Latent Spectral Dynamics, a framework that reconceptualizes autoregressive LLM generation as a stochastic, autonomous dynamical system, and extracts its governing transition operators via time-delay embedding and Hankel-EDMD approximation of the Koopman operator. The empirical consequences suggest that truthful and untruthful generations share a stable latent backbone yet diverge in temporally revealed oscillatory Koopman modes — undetectable from static snapshots, robust across sample sizes, generalizable without supervision to an unlabeled domain — establishing a concrete spectral marker of epistemic states whose conserved frequency across model scales suggests a broader dynamical signature of model uncertainty. Characterizing this connection, extending to longer generation contexts, and moving beyond a single model family and the linear Koopman approximation are important directions for future work. On the methodological side, learnable neural Koopman dictionaries could replace the current Hankel-EDMD pipeline, bypassing its cubic scaling and regularization requirements while potentially capturing nonlinear observable structure that coordinate bases miss. More broadly, the operator-theoretic lens is not intrinsically limited to hallucination — strategic deception, persona drift, and sycophancy each involve characteristic temporal signatures that may admit analogous spectral diagnoses. We hope this work establishes a foundation for moving from static dissection of model internals toward a *temporal interpretability*: a principled science of generative behavior at the activation-trajectory level.

7 LLM Usage Statement

Large language models (Gemini, Claude, ChatGPT) were used to assist with idea exploration, theorem development, conducting experiments, graphics generation, and paper writing and editing.

References

Mohamed Akrout. Representations matter: Embedding modes of large language models using dynamic mode decomposition. *arXiv preprint arXiv:2309.01245*, 2023.

Shaojie Bai, J Zico Kolter, and Vladlen Koltun. Deep equilibrium models. *Advances in neural information processing systems*, 32, 2019.

Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.

Jonah Botvinick-Greenhouse, Maria Oprea, Romit Maulik, and Yunan Yang. Measure-theoretic time-delay embedding. *Journal of Statistical Physics*, 2025. arXiv:2409.08768.

Ricky TQ Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. *Advances in neural information processing systems*, 31, 2018.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 1(1):12, 2021.

Mingmeng Geng, Amr Mohamed, Guokan Shang, Michalis Vazirgiannis, and Thierry Poibeau. Markovian generation chains in large language models. *arXiv preprint arXiv:2603.11228*, 2026.

Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. *arXiv preprint arXiv:2401.06102*, 2024.

Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, et al. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*, 2024.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2024.

Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Lanham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, et al. Sleeper agents: Training deceptive LLMs that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024.

Nandan Kumar Jha and Brandon Reagen. NerVE: Nonlinear eigenspectrum dynamics in LLM feed-forward networks. In *International Conference on Learning Representations (ICLR)*, 2026.

Bernard O Koopman. Hamiltonian systems and transformation in hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.

JinXin Li, Gang Tu, and JunJie Hu. Llm hallucination detection: Hsad. *arXiv preprint arXiv:2509.23580*, 2025a.

Zhaoyang Li, Minghao Han, and Xunyuan Yin. Mamko: Mamba-based koopman operator for modeling and predictive control. In *ICLR*, 2025b.

Igor Mezić. Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41(1):309–325, 2005.

Usman Naseem. Mechanistic interpretability for large language model alignment: Progress, challenges, and future directions. *arXiv preprint arXiv:2602.11180*, 2026.

Valentin Noël. Geometry of reason: Spectral signatures of valid mathematical reasoning. *arXiv preprint arXiv:2601.00791*, 2026.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint*, 2023.

Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. *arXiv preprint*, 2024.

Peter J Schmid. Dynamic mode decomposition of numerical and experimental data. *Journal of fluid mechanics*, 656:5–28, 2010.

S. M. Tonmoy, S. M. M. Zaman, Vinija Jain, Anku Rani, Vipula Rawte, Aman Chadha, and Amitava Das. A comprehensive survey of hallucination mitigation techniques in large language models. *arXiv preprint arXiv:2401.01313*, 2024.

Haokun Wang et al. Tracking the feature dynamics in LLM training: A mechanistic study. *arXiv preprint arXiv:2412.17626*, 2024.

Matthew O Williams, Ioannis G Kevrekidis, and Clarence W Rowley. A data–driven approximation of the koopman operator: Extending dynamic mode decomposition. *Journal of Nonlinear Science*, 25(6):1307–1346, 2015.

Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit bayesian inference. *arXiv preprint arXiv:2111.02080*, 2021.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Oussama Zekri, Ambroise Odonnat, Abdelhakim Benechehab, Linus Bleistein, Nicolas Boullé, and Ievgen Redko. Large language models as markov chains. *arXiv preprint arXiv:2410.02724*, 2024.

Shimin Zhang, Xianwei Chen, Yufan Shen, Ziyuan Ye, and Jibin Wu. Relax: Reasoning with latent exploration for large reasoning models. *arXiv preprint arXiv:2512.07558*, 2025a.

Shimin Zhang, Ziyuan Ye, Yinsong Yan, Zeyang Song, Yujie Wu, and Jibin Wu. Koopstd: Reliable similarity analysis between dynamical systems via approximating koopman spectrum with timescale decoupling. In *Forty-second International Conference on Machine Learning*, 2025b.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023a.

Andy Zou, Zifan Wang, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023b.

A Theoretical Proofs and Derivations

This appendix provides the mathematical derivations supporting the dynamical framework introduced in Sections 3.1–3.4 of the main text.

A.1 The KV-Cache Markov Chain and Autonomous Formulation

Let ϕ denote the frozen parameters of an autoregressive Transformer with L layers, residual stream dimension d, n_{heads} attention heads of dimension d_{head}, and vocabulary $\mathcal{V}$. At generation step k, the KV cache is $\mathcal{S}_k = \{(K_\ell^{(j)}, V_\ell^{(j)})\}_{\ell=1,\, j=1}^{L,\, p+k}$, where p is the prompt length.

Proposition 1 (Exact Markov Property) *The sequence $\{\mathcal{S}_k\}_{k=0}^{L_{\max}}$ is a Markov chain: for all k and all measurable A,*

$$P_\phi(\mathcal{S}_{k+1} \in A \mid \mathcal{S}_0, \ldots, \mathcal{S}_k) = P_\phi(\mathcal{S}_{k+1} \in A \mid \mathcal{S}_k).$$

Proof: The cache update $\mathcal{S}_{k+1} = U_\phi(\mathcal{S}_k, x_{k+1})$ is deterministic given $(\mathcal{S}_k, x_{k+1})$. A deterministic forward pass produces $h_k = f_\phi^{(L)}(\mathcal{S}_k)$, and the next token is sampled from $P_\phi(x_{k+1} = v \mid \mathcal{S}_k) = \text{Softmax}_v(l(h_k)/\tau)$, where $l(h_k) = W_U h_k + b_U$. Since x_{k+1} is conditionally independent of $\mathcal{S}_0, \ldots, \mathcal{S}_{k-1}$ given $\mathcal{S}_k$, and $\mathcal{S}_{k+1}$ is a deterministic function of $(\mathcal{S}_k, x_{k+1})$, the Markov property follows. □

Autonomous system on $\{\mathcal{S}_k\}$. To express the cache dynamics as a continuous-state stochastic system, we apply the Gumbel-Max reparameterization. Let $\Xi_k \in \mathbb{R}^{|\mathcal{V}|}$ be a vector of i.i.d. standard Gumbel random variables. The sampled token becomes a deterministic function of $(\mathcal{S}_k, \Xi_k)$:

$$x_{k+1} = \arg\max_{v \in \mathcal{V}} \left(\frac{l(h_k)_v}{\tau} + \Xi_{k,v} \right) \equiv \mathcal{T}_{\tau,\phi}(\mathcal{S}_k, \Xi_k) \tag{10}$$

where $h_k = f_\phi^{(L)}(\mathcal{S}_k)$ and $l(h_k) = W_U h_k + b_U$. Substituting into the cache update eliminates x_{k+1}:

$$\mathcal{S}_{k+1} = U_\phi(\mathcal{S}_k,\ \mathcal{T}_{\tau,\phi}(\mathcal{S}_k, \Xi_k)) := F_\phi(\mathcal{S}_k, \Xi_k) \tag{11}$$

Because F_ϕ is deterministic in both arguments and Ξ_k is exogenous i.i.d. noise, $\{\mathcal{S}_k\}$ is an autonomous, discrete-time stochastic dynamical system on $\mathcal{M}$.

A.2 Compactness of the State Space and Observation Map Regularity

This section establishes the two prerequisites invoked by the delay embedding guarantee in §3.2.2: compactness of the reachable state space and Lipschitz regularity of the observation map.

A.2.1 Compact state space

Proposition 2 (Compact State Space) *For any fixed autoregressive Transformer with frozen parameters ϕ, maximum generation length $L_{\max}$, and prompt length p, the set of reachable KV cache states across all generation steps $k \leq L_{\max}$ lies within a compact subset of a fixed Euclidean space.*

Proof: We proceed in three steps: embedding into a fixed ambient space, establishing uniform boundedness, and proving compactness.

Step 1: Fixed dimensionality. The number of KV positions never exceeds $p + L_{\max}$. We embed every cache state $\mathcal{S}_k$ into a fixed ambient space by zero-padding unoccupied positions, yielding $\widetilde{\mathcal{S}}_k = [\mathcal{S}_k, \mathbf{0}] \in \mathbb{R}^{D_{\max}}$, where

$$D_{\max} = (p + L_{\max}) \cdot 2L \cdot n_{\text{heads}} \cdot d_{\text{head}}.$$

Step 2: Uniform boundedness. Every KV entry is a linear projection of a LayerNorm output: $K_\ell^{(j)} = W_K^{(\ell)}\, \text{LayerNorm}(z_\ell^{(j)})$. LayerNorm normalizes its input to zero mean and unit variance, then applies learned affine parameters $\gamma^{(\ell)}, \beta^{(\ell)}$. For any input $z \in \mathbb{R}^d$:

$$\|\,\text{LayerNorm}(z)\|_2 \;\leq\; \|\gamma^{(\ell)}\|_2 + \sqrt{d}\,\|\beta^{(\ell)}\|_\infty \;=:\; B_{\text{LN}}^{(\ell)}.$$

This bound is independent of z, k, and the sequence history. Each KV entry therefore satisfies $\|K_\ell^{(j)}\|_2 \leq \|W_K^{(\ell)}\|_{\text{op}} B_{\text{LN}}^{(\ell)}$, and analogously for $V_\ell^{(j)}$. Summing over all layers, heads, and positions, the full zero-padded cache satisfies

$$\|\widetilde{\mathcal{S}}_k\|_2 \;\leq\; R(\phi)$$

for a finite constant $R(\phi)$ depending only on the frozen weights. In particular, every reachable state lies in the closed ball $\overline{B}(0, R(\phi)) \subset \mathbb{R}^{D_{\max}}$.

Step 3: Compactness. Let $\mathcal{R} = \{\widetilde{\mathcal{S}}_k : k \leq L_{\max},\ \text{trajectory} \sim \mu\}$ denote the set of all reachable zero-padded cache states. By Step 2, $\mathcal{R} \subseteq \overline{B}(0, R(\phi))$ is bounded. Define $\mathcal{M} = \overline{\mathcal{R}}$, the closure of $\mathcal{R}$ in $\mathbb{R}^{D_{\max}}$. Then $\mathcal{M}$ is closed by construction. Since the closure of a bounded set is bounded ($\mathcal{R} \subseteq \overline{B}(0, R(\phi))$ implies $\overline{\mathcal{R}} \subseteq \overline{B}(0, R(\phi))$, as the closed ball is itself closed), $\mathcal{M}$ is a closed, bounded subset of $\mathbb{R}^{D_{\max}}$. By the Heine–Borel theorem, $\mathcal{M}$ is compact.

For the sub-manifold $\mathcal{M}_{\theta^*}$ of Assumption 1, we take its closure $\overline{\mathcal{M}_{\theta^*}}$ within $\mathcal{M}$. Since $\overline{\mathcal{M}_{\theta^*}}$ is a closed subset of the compact set $\mathcal{M}$, it is itself compact. The delay embedding analysis operates on this closure; we continue to write $\mathcal{M}_{\theta^*}$ for notational convenience. □

A.2.2 Observation map regularity

Proposition 3 (Observation Map Regularity) *The observation map $g : \mathcal{M} \to \mathbb{R}^d$ defined by $g(\widetilde{\mathcal{S}}_k) = h_k = f_\phi^{(L)}(\widetilde{\mathcal{S}}_k)$ is Lipschitz continuous on $\mathcal{M}$.*

Proof: The map g is a finite composition of the following operations, each of which is Lipschitz on the compact domain $\mathcal{M}$:

1. *Linear projections* ($W_Q, W_K, W_V, W_O, W_1, W_2$): globally Lipschitz with constant equal to the operator norm.
2. *Softmax attention*: the softmax function $\sigma : \mathbb{R}^n \to \Delta^{n-1}$ is smooth on all of $\mathbb{R}^n$ and hence Lipschitz on any compact subset. The attention map $\text{Attn}(Q, K, V) = \sigma(QK^\top / \sqrt{d_{\text{head}}})\, V$ is a composition of bilinear maps (Lipschitz on bounded domains) and softmax, hence Lipschitz on $\mathcal{M}$.
3. LayerNorm: with the standard implementation using $\epsilon > 0$ in the denominator, LayerNorm is smooth on all of $\mathbb{R}^d$ (the denominator $\sqrt{\text{Var}(z) + \epsilon}$ is bounded away from zero) and hence Lipschitz on $\mathcal{M}$.
4. *Activation functions* (GeLU or SiLU): both are smooth with globally bounded derivatives, hence globally Lipschitz.
5. *Residual connections*: linear maps (addition), Lipschitz with constant 1.

The composition of finitely many Lipschitz maps is Lipschitz, with constant bounded by the product $\prod_{i=1}^n L_i$ of the individual Lipschitz constants. □

A.3 Delay Embedding: Guarantees and Empirical Validation

This section provides the formal justification for the delay embedding construction in §3.2, expanding on the theorem statement and assumptions summarized in the main text.

A.3.1 The observation bottleneck

The observation map $g : \mathcal{M} \to \mathbb{R}^d$ (Proposition 3) projects the high-dimensional cache state $\widetilde{\mathcal{S}}_k \in \mathcal{M}$ into the d-dimensional residual stream h_k. Because the logit vector is $l(h_k) = W_U h_k + b_U$, a linear function of h_k alone, the generation process factors through h_k: the distribution of x_{k+1} depends on $\mathcal{S}_k$ only through h_k, so h_k is an exact sufficient statistic for the immediate next token. However, computing h_{k+1} requires attending to the full cache $\mathcal{S}_{k+1}$, which retains information from $\mathcal{S}_k$ that h_k discards. The observable sequence $\{h_k\}$ is therefore non-Markovian, motivating the delay construction.

A.3.2 Delay-coordinate map

Since autoregressive generation at $\tau > 0$ is stochastic, past states $\mathcal{S}_{k-j}$ are not deterministic functions of the current state $\mathcal{S}_k$ (different noise realizations produce different histories for the same current state). The delay-coordinate map therefore acts on *trajectory segments*:

$$\Phi_W : \mathcal{M}^W \to \mathbb{R}^{Wd}, \qquad \Phi_W(\mathcal{S}_{k-W+1}, \ldots, \mathcal{S}_k) = \left[g(\mathcal{S}_{k-W+1})^\top, \ldots, g(\mathcal{S}_k)^\top\right]^\top = \mathbf{H}_k \quad (12)$$

At $\tau = 0$ (greedy decoding), the dynamics are deterministic, so $\mathcal{S}_{k-j} = F_\phi^{-j}(\mathcal{S}_k)$ on the attractor, and Φ_W reduces to a function of $\mathcal{S}_k$ alone—recovering the classical Takens setting.

A.3.3 Low-dimensional concentration

Under a fixed prompting regime θ^*, generation trajectories concentrate on a sub-manifold $\mathcal{M}_{\theta^*} \subseteq \mathcal{M}$ of intrinsic dimension $m_{\text{eff}} \ll D_{\max}$ (Assumption 1). Two structural properties support this assumption:

1. *Bounded generative entropy.* Under directive prompts, the next-token distribution concentrates on a small effective support at each step, constraining the degrees of freedom of the trajectory distribution.
2. *Attention locality.* Rotary Position Embedding (RoPE) induces attention decay with positional distance, confining the dynamically relevant portion of the cache to a neighborhood of recent tokens.

A.3.4 Formal embedding guarantee

With compactness (Proposition 2) and low-dimensional concentration (Assumption 1) established, we state the embedding theorem that underpins the approximation in Eq. (3).

Theorem 1 (Measure-Theoretic Time-Delay Embedding) *Let $\mathcal{M}_{\theta^*} \subseteq \mathcal{M}$ be a compact sub-manifold of intrinsic dimension m_{eff}, and let $g : \mathcal{M} \to \mathbb{R}^d$ be a smooth (C^r, $r \geq 1$) observation map. For generic g (in the sense of a residual set in the C^r topology) and $W \geq 2\, m_{\text{eff}} + 1$, the delay-coordinate map Φ_W preserves the invariant measure of the dynamics under pushforward: the pushforward $(\Phi_W)_\# \mu$ of the invariant measure μ on trajectory segments uniquely determines μ (Botvinick-Greenhouse et al., 2025).*

Consequence for the Koopman operator. Because measure preservation places $L^2(\mu)$ functions on $\mathcal{M}_{\theta^*}$ in bijective correspondence with $L^2((\Phi_W)_\# \mu)$ functions on $\mathbb{R}^{Wd}$, the Koopman operator estimated from $\{\mathbf{H}_k\}$ inherits the spectral properties of the true operator on $\{\mathcal{S}_k\}$. Since the next-token distribution $P_\phi(x_{k+1} = v \mid \mathcal{S}_k)$ is a bounded functional of $\mathcal{S}_k$ (hence an element of $L^2(\mu)$), this establishes

$$P_\phi(x_{k+1} \mid \mathcal{S}_k) \;\approx\; P_\phi(x_{k+1} \mid \mathbf{H}_k) \quad (13)$$

in the $L^2(\mu)$ sense, justifying the approximation stated in Eq. (3).

A.3.5 Scope of the theoretical guarantee

Theorem 1 provides the mathematical foundation for the delay-embedding strategy: it establishes that faithful phase-space reconstruction *is possible in principle* from delay-embedded observations. However, two of its conditions are not formally verified in our experimental setting, and we document these gaps explicitly.

1. **Regularity and genericity of the observation map.** The theorem requires a smooth (C^r) observation map belonging to a generic (residual) set in the C^r topology. By Proposition 3, our observation map $g = f_\phi^{(L)}$ is Lipschitz continuous on $\mathcal{M}$. It is additionally smooth almost everywhere: the only source of non-smoothness is the softmax attention function at points where two or more attention logits are exactly equal—a measure-zero set under any continuous input distribution. While Lipschitz regularity and almost-everywhere smoothness are weaker than the

global C^r smoothness required by the theorem, two considerations mitigate this gap: the non-smooth set has measure zero under the generation distribution and therefore does not affect the $L^2(\mu)$ spectral analysis; and the high dimensionality of h_k ($d = 5120$) makes pathological alignment between the observation map and the kernel of Φ_W extremely unlikely.
Furthermore, the genericity condition asserts that the theorem holds for a residual set of observation maps. Our map g is a specific, fixed function determined by the Transformer weights—not a generic element of this set. We cannot formally verify membership in the generic set without constructing an explicit perturbation argument.

2. **Window size.** With $m_{\text{eff}} \approx 30$, the formal bound requires $W \geq 61$. Our experiments use $W \in \{1, 3, 5, 8\}$, well below this threshold. Section A.3.6 discusses the structural properties that mitigate this gap, and Section A.3.7 introduces the empirical criterion that validates spectral recovery at practical window sizes.

In summary, Theorem 1 provides the *theoretical motivation* for the delay-embedding approach—establishing that Koopman spectral recovery from observations is mathematically grounded—while the ε-predictive criterion (Definition 1) provides the *empirical validation* that the dominant spectral modes are faithfully recovered in our specific setting.

A.3.6 The window-size gap

Two structural properties of the Transformer architecture provide reason to expect faithful spectral recovery at window sizes smaller than the formal bound $2\, m_{\text{eff}} + 1$:

1. **Structured observation.** The Transformer's observation map is not a generic low-dimensional projection: multi-head attention aggregates information across the full KV cache at each step, concentrating predictive information in h_k far beyond what a random d-dimensional projection would achieve. This built-in state compression can enable faithful reconstruction at smaller W than the worst-case generic bound requires.
2. **Spectral recovery $\neq$ full embedding.** Accurate Koopman spectral analysis does not require global injectivity of Φ_W or full measure preservation. It requires the weaker condition that the Koopman eigenfunctions of interest project nontrivially onto the observation space—i.e., that they are not orthogonal to the span of the chosen observables. The dominant modes (those with $|\lambda| \approx 1$) capture slowly varying, large-scale structure that is visible in any reasonable observation of the state.

A.3.7 Empirical validation: the ε-predictive criterion

To bridge the gap between formal guarantees and practical performance, we employ the ε-predictive embedding criterion introduced in Definition 1. The delay map Φ_W is an ε-predictive embedding if the estimated Koopman operator $\mathbf{K}$ achieves held-out relative one-step MSE at most ε:

$$\text{Rel-MSE}(1) \;:=\; \frac{\sum_i \|\mathbf{H}_{k_i+1} - \mathbf{H}_{k_i}\mathbf{K}\|_2^2}{\sum_i \|\mathbf{H}_{k_i+1}\|_2^2} \;\leq\; \varepsilon \tag{14}$$

where the sum runs over held-out test transitions (disjoint from the training set used to estimate $\mathbf{K}$).

An ε-predictive embedding guarantees that the operator captures at least fraction $(1 - \varepsilon)$ of the predictable transition variance. Because the dominant Koopman eigenvalues govern the largest-variance modes of the transition, this ensures they are well-estimated even when the formal embedding conditions of Theorem 1 are not fully satisfied.

In §5.1, the coordinate-basis operator achieves $\varepsilon \approx 0.17$–0.31 across window sizes and classes, with stable autonomous rollouts (no ground-truth injection) through horizon $h = 8$. This multi-step stability provides strong evidence that the linear operator captures the dominant dynamical structure: errors in the estimated eigenvalues would compound multiplicatively under rollout, causing rapid divergence at horizons well before $h = 8$.

A.4 Derivation of the Autonomous Dynamics on $\{\mathbf{H}_k\}$

This section derives the autonomous transition map $\mathcal{F}_{\phi,\theta^*}$ stated in main-text Eq. (6), bridging the Gumbel-Max reparameterization with the delay embedding approximation.

Gumbel-Max reparameterization. Standard autoregressive generation samples x_{k+1} from a categorical distribution parameterized by the temperature-scaled softmax:

$$P_\phi(x_{k+1} = v \mid \mathbf{H}_k) = \mathrm{Softmax}_v\left(\frac{l(h_k)}{\tau}\right) = \frac{\exp(l(h_k)_v/\tau)}{\sum_{j\in\mathcal{V}} \exp(l(h_k)_j/\tau)} \tag{15}$$

where h_k is the most recent component of $\mathbf{H}_k$ and $l(h_k) = W_U h_k + b_U \in \mathbb{R}^{|\mathcal{V}|}$. Let $\Xi_k \in \mathbb{R}^{|\mathcal{V}|}$ be i.i.d. standard Gumbel noise (realized via $\Xi_{k,v} = -\log(-\log(U_{k,v}))$ for $U_{k,v} \sim \mathrm{Uniform}(0,1)$). The Gumbel-Max theorem gives:

$$x_{k+1} = \arg\max_{v\in\mathcal{V}}\left(\frac{l(h_k)_v}{\tau} + \Xi_{k,v}\right) \equiv \mathcal{T}_{\tau,\phi}(\mathbf{H}_k, \Xi_k) \tag{16}$$

This decouples the categorical stochasticity: token generation is a deterministic function $\mathcal{T}_{\tau,\phi}$ of continuous inputs.

Hankel window update. Upon generating x_{k+1}, the Transformer performs a deterministic forward pass on the updated cache to compute h_{k+1}. The Hankel window shifts forward, discarding h_{k-W+1} and appending h_{k+1}:

$$\mathbf{H}_{k+1} = \mathcal{U}_\phi(\mathbf{H}_k, x_{k+1}) = \left[h_{k-W+2}^\top, \ldots, h_k^\top,\ h_{k+1}^\top\right]^\top \tag{17}$$

where h_{k+1} depends on the full updated cache $\mathcal{S}_{k+1} = U_\phi(\mathcal{S}_k, x_{k+1})$.

Closing the loop via the delay embedding. The update (17) depends on $\mathcal{S}_k$ through the forward pass that produces h_{k+1}. Under the delay embedding approximation of §3.2, $\mathbf{H}_k$ approximately determines $\mathcal{S}_k$ (in the $L^2(\mu)$ sense established in Appendix A.3). Substituting $\mathcal{T}_{\tau,\phi}$ into $\mathcal{U}_\phi$ and invoking this approximation yields the autonomous system:

$$\mathbf{H}_{k+1} = \mathcal{U}_\phi\big(\mathbf{H}_k,\ \mathcal{T}_{\tau,\phi}(\mathbf{H}_k, \Xi_k)\big)\ :=\ \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \Xi_k) \tag{18}$$

The map $\mathcal{F}_{\phi,\theta^*}$ is deterministic in both $\mathbf{H}_k$ and Ξ_k. Since Ξ_k is exogenous i.i.d. noise, the sequence $\{\mathbf{H}_k\}$ evolves as an approximate Markov chain, governed by the transition kernel:

$$\mathcal{P}_{\theta^*}(d\mathbf{H}_{k+1} \mid \mathbf{H}_k) = \mathbb{P}_\Xi\big(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \Xi_k) \in d\mathbf{H}_{k+1}\big) \tag{19}$$

The subscript θ^* indexes the prompting regime: distinct regimes produce distinct training trajectories and hence distinct operators. As $\tau \to 0^+$, Ξ_k becomes irrelevant and the system is purely deterministic: $\mathbf{H}_{k+1} = \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k)$.

A.5 Background: Koopman Operator Theory

This appendix provides the theoretical background for the Koopman operator framework introduced in §3.4, details the finite-dimensional approximation procedure, and proves the spectral equivalence between the primal and dual formulations.

A.5.1 The Koopman Operator on Stochastic Systems

Let $(\mathbb{R}^{Wd}, \mathcal{B}, \mu)$ be the measure space defined by the Borel σ-algebra on the Hankel state space equipped with the invariant measure μ of the Markov chain $\{\mathbf{H}_k\}$ established in §3.3. The *Koopman operator* $\mathcal{K}_{\theta^*} : L^2(\mu) \to L^2(\mu)$ associated with the transition kernel $\mathcal{P}_{\theta^*}$ is defined by

$$(\mathcal{K}_{\theta^*} g)(\mathbf{H}_k) = \mathbb{E}_{\mathbf{H}_{k+1}\sim\mathcal{P}_{\theta^*}(\cdot\mid\mathbf{H}_k)}\left[g(\mathbf{H}_{k+1})\right] = \mathbb{E}_{\Xi_k}\left[g\big(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k, \Xi_k)\big)\right] \tag{20}$$

for any observable $g \in L^2(\mu)$, where the second equality follows from the autonomous formulation (Eq. 7) and the exogeneity of Ξ_k.

Several properties of $\mathcal{K}_{\theta^*}$ are immediate from this definition and relevant to our analysis:

Linearity. Although the transition map $\mathcal{F}_{\phi,\theta^*}$ is nonlinear, $\mathcal{K}_{\theta^*}$ acts *linearly* on observables: for any $g_1, g_2 \in L^2(\mu)$ and $\alpha, \beta \in \mathbb{R}$,

$$\mathcal{K}_{\theta^*}(\alpha g_1 + \beta g_2) = \alpha\, \mathcal{K}_{\theta^*} g_1 + \beta\, \mathcal{K}_{\theta^*} g_2. \tag{21}$$

This linearity is the central advantage of the Koopman framework: it recasts nonlinear state-space dynamics as linear evolution in observable space, making spectral decomposition applicable to inherently nonlinear systems.

Boundedness. Since $\mathcal{P}_{\theta^*}$ is a Markov kernel, Jensen's inequality gives $\|\mathcal{K}_{\theta^*} g\|_{L^2(\mu)} \leq \|g\|_{L^2(\mu)}$ for all $g \in L^2(\mu)$, so $\mathcal{K}_{\theta^*}$ is a contraction (operator norm ≤ 1). In particular, all eigenvalues satisfy $|\lambda| \leq 1$.

Spectral decomposition. If $\mathcal{K}_{\theta^*}$ admits eigenvalues $\{\lambda_j\}_{j=1}^{\infty}$ with corresponding eigenfunctions $\{\varphi_j\}_{j=1}^{\infty} \subset L^2(\mu)$, then $\mathcal{K}_{\theta^*}\varphi_j = \lambda_j \varphi_j$ and any observable $g \in L^2(\mu)$ in the closed span of the eigenfunctions can be decomposed as

$$g(\mathbf{H}_k) = \sum_{j=1}^{\infty} \varphi_j(\mathbf{H}_0)\, \lambda_j^k\, v_j, \tag{22}$$

where v_j are the *Koopman modes* obtained by projecting g onto the eigenfunctions. This decomposition separates the dynamics into independently evolving components: each eigenvalue λ_j determines the temporal behavior of its associated mode (decay rate $|\lambda_j|$, oscillation frequency $\arg(\lambda_j)$).

A.5.2 Deterministic Limit

As the sampling temperature $\tau \to 0^+$, the Gumbel noise Ξ_k becomes irrelevant and token selection becomes deterministic. Let $v^* = \arg\max_v \ell(\mathbf{h}_k)_v$ (assumed unique; ties form a measure-zero set under any continuous logit distribution) and $\Delta_v := \ell(\mathbf{h}_k)_{v^*} - \ell(\mathbf{h}_k)_v > 0$ for $v \neq v^*$. A competing token v wins the arg max in Eq. (6) only if $\Xi_{k,v} - \Xi_{k,v^*} > \Delta_v/\tau$. Since $\Xi_{k,v} - \Xi_{k,v^*} \sim \mathrm{Logistic}(0,1)$,

$$\mathbb{P}[\![\,(x_{k+1} \neq v^* \mid \mathbf{H}_k) \;=\; 1 - \frac{1}{1 + \sum_{v \neq v^*} e^{-\Delta_v/\tau}} \;=\; \frac{\sum_{v \neq v^*} e^{-\Delta_v/\tau}}{1 + \sum_{v \neq v^*} e^{-\Delta_v/\tau}} \;\leq\; \sum_{v \neq v^*} e^{-\Delta_v/\tau} \;\xrightarrow{\tau \to 0^+}\; 0, \tag{23}$$

so $x_{k+1} \to v^*$ almost surely and the transition map becomes $\mathbf{H}_{k+1} = \mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k)$. The Markov kernel collapses to the Dirac measure $\mathcal{P}_{\theta^*}(A \mid \mathbf{H}_k) = \mathbf{1}[\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k) \in A]$, and substituting into Eq. (20) yields the classical composition operator:

$$(\mathcal{K}_{\theta^*} g)(\mathbf{H}_k) = \int g(\mathbf{H})\, \delta_{\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k)}(d\mathbf{H}) = g\big(\mathcal{F}_{\phi,\theta^*}(\mathbf{H}_k)\big), \tag{24}$$

recovering the deterministic Koopman operator (Koopman, 1931). The stochastic formulation in Eq. (20) generalizes it to $\tau > 0$ by replacing pointwise composition with conditional expectation, following Mezić (2005).

A.5.3 Finite-Dimensional Approximation via EDMD

The infinite-dimensional operator $\mathcal{K}_{\theta^*}$ cannot be computed directly. Extended Dynamic Mode Decomposition (EDMD) (Williams et al., 2015) approximates it by projecting onto a finite-dimensional subspace spanned by a chosen dictionary of observables $\{g_1, \ldots, g_D\}$.

Dictionary evaluation. Given N transition pairs $\{(\mathbf{H}_k, \mathbf{H}_{k+1})\}$ collected from generation trajectories, we evaluate the dictionary at each state to form data matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{N \times D}$:

$$\mathbf{X} = \begin{bmatrix} g(\mathbf{H}_0)^\top \\ \vdots \\ g(\mathbf{H}_{N-1})^\top \end{bmatrix}, \qquad \mathbf{Y} = \begin{bmatrix} g(\mathbf{H}_1)^\top \\ \vdots \\ g(\mathbf{H}_N)^\top \end{bmatrix}, \tag{25}$$

where $g(\mathbf{H}_k) = [g_1(\mathbf{H}_k), \ldots, g_D(\mathbf{H}_k)]^\top \in \mathbb{R}^D$. For the coordinate basis used in our main results, $g(\mathbf{H}_k) = \mathbf{H}_k$ and $D = Wd$.

Galerkin projection. We seek a finite-rank operator $\widehat{\mathcal{K}}$ acting within $\mathrm{span}\{g_1, \ldots, g_D\}$ that satisfies the Galerkin condition $\langle g_i, \mathcal{K}_{\theta^*} g_j - \widehat{\mathcal{K}} g_j \rangle_{L^2(\mu)} = 0$ for all i, j. Writing $\widehat{\mathcal{K}} g_j = \sum_m K_{mj}\, g_m$ gives the matrix equation $\mathbf{G}\,\mathbf{K} = \mathbf{A}$ with

$$G_{ij} = \mathbb{E}_\mu\big[g_i(\mathbf{H}_k)\, g_j(\mathbf{H}_k)\big], \qquad A_{ij} = \mathbb{E}_\mu\big[g_i(\mathbf{H}_k)\, g_j(\mathbf{H}_{k+1})\big]. \tag{26}$$

The data matrices provide empirical estimates $\widehat{\mathbf{G}} = \frac{1}{N}\mathbf{X}^\top\mathbf{X}$ and $\widehat{\mathbf{A}} = \frac{1}{N}\mathbf{X}^\top\mathbf{Y}$, reducing the Galerkin equation to $\mathbf{X}^\top\mathbf{X}\,\mathbf{K} = \mathbf{X}^\top\mathbf{Y}$.

Tikhonov-regularized least-squares objective. Equivalently, the EDMD operator is obtained by solving the regularized regression

$$\mathbf{K}_{\text{primal}} = \underset{\mathbf{K} \in \mathbb{R}^{D \times D}}{\arg\min}\ \frac{1}{2}\,\|\mathbf{Y} - \mathbf{X}\mathbf{K}\|_F^2 + \frac{\alpha}{2}\,\|\mathbf{K}\|_F^2. \tag{27}$$

Differentiating with respect to $\mathbf{K}$ and setting the result to zero:

$$\frac{\partial J}{\partial \mathbf{K}} = \big(\mathbf{X}^\top\mathbf{X} + \alpha\,\mathbf{I}_D\big)\mathbf{K} - \mathbf{X}^\top\mathbf{Y} = \mathbf{0}, \tag{28}$$

which, since $\mathbf{X}^\top\mathbf{X} + \alpha\,\mathbf{I}_D \succ 0$ for any $\alpha > 0$, admits the unique solution

$$\mathbf{K}_{\text{primal}} = \big(\mathbf{X}^\top\mathbf{X} + \alpha\,\mathbf{I}_D\big)^{-1}\mathbf{X}^\top\mathbf{Y}. \tag{29}$$

Connection to Vector Autoregression. For the coordinate basis $g(\mathbf{H}) = \mathbf{H}$, the EDMD objective (27) reduces to a Tikhonov-regularized multivariate regression of $\mathbf{H}_{k+1}$ on $\mathbf{H}_k$. Since $\mathbf{H}_k$ contains W consecutive residual stream vectors, this is equivalent to a regularized Vector Autoregression of order W (VAR(W)) on the residual stream sequence $\{\mathbf{h}_k\}$, with $\mathbf{K}_{\text{primal}}$ playing the role of the VAR companion matrix. The spectral analysis of $\mathbf{K}_{\text{primal}}$ therefore extracts the same dynamical modes as a VAR analysis, but the Koopman framing situates them within the broader operator-theoretic framework, connecting finite-sample estimates to the infinite-dimensional spectral theory.

Computational cost. The dual formulation reduces eigendecomposition from $O(D^3)$ to $O(N^3)$. In our setting this is a reduction from $O(25{,}600^3) \approx 10^{13}$ to $O(3{,}000^3) \approx 10^{10}$ — a factor of $\sim 10^3$. The Gram matrix $\mathbf{X}\mathbf{X}^\top \in \mathbb{R}^{N \times N}$ and the product $\mathbf{Y}\mathbf{X}^\top$ are computed once and reused across all regularization parameters during grid search.

A.5.4 Tikhonov Regularization and Temperature Effects

The regularization parameter α controls the bias–variance tradeoff of the estimated operator. In the dual formulation, α appears as a ridge penalty on the Gram matrix $\mathbf{X}\mathbf{X}^\top + \alpha\,\mathbf{I}_N$, shrinking all eigenvalues of the Koopman estimate toward zero.

Role of α. Small α yields an operator that closely fits the training transitions but may overfit to token-level noise, producing spurious eigenvalues near the unit circle. Large α over-regularizes, collapsing the spectrum toward the origin and suppressing genuine dynamical modes. We select α via grid search over $\alpha \in [10^3, 10^4]$, choosing the value that minimizes held-out one-step relative MSE (Definition 1).

Temperature-induced spectral scattering. At higher sampling temperatures τ, the Gumbel noise Ξ_k contributes greater variance to each transition. This manifests as increased scatter in the estimated Koopman spectrum: eigenvalues that are tightly clustered at $\tau = 0$ spread outward at $\tau = 0.3$ and further at $\tau = 0.7$. Tikhonov regularization counteracts this effect by damping the high-variance spectral components that arise from stochastic forcing, effectively recovering the deterministic backbone spectrum. The regularization parameter α must therefore be adapted to the temperature: higher τ requires larger α to maintain spectral stability, a relationship we observe empirically in the grid search.

A.5.5 Dual Formulation: Exact Spectral Equivalence

Theorem 2 (Spectral Equivalence) *The non-zero eigenvalues of the high-dimensional primal Koopman operator $\mathbf{K}_{\text{primal}} \in \mathbb{R}^{D\times D}$ are exactly equal to the eigenvalues of the reduced dual operator $\mathbf{K}_{\text{dual}} \in \mathbb{R}^{N\times N}$, linked by the lifting equation $\mathbf{v}_{\text{primal}} = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}\mathbf{v}_{\text{dual}}$.*

Proof: The Tikhonov-regularized least-squares objective for the primal operator is:

$$\min_{\mathbf{K}_{\text{primal}}} \frac{1}{2}\|\mathbf{Y} - \mathbf{X}\mathbf{K}_{\text{primal}}\|_F^2 + \frac{\alpha}{2}\|\mathbf{K}_{\text{primal}}\|_F^2 \tag{30}$$

Setting the derivative to zero yields the normal equation:

$$(\mathbf{X}^\top\mathbf{X} + \alpha\mathbf{I}_D)\,\mathbf{K}_{\text{primal}} = \mathbf{X}^\top\mathbf{Y} \implies \mathbf{K}_{\text{primal}} = (\mathbf{X}^\top\mathbf{X} + \alpha\mathbf{I}_D)^{-1}\mathbf{X}^\top\mathbf{Y} \tag{31}$$

Since $D \gg N$, we apply the push-through identity:

$$(\mathbf{X}^\top\mathbf{X} + \alpha\mathbf{I}_D)^{-1}\mathbf{X}^\top = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1} \tag{32}$$

yielding:

$$\mathbf{K}_{\text{primal}} = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}\mathbf{Y} \tag{33}$$

Define the dual operator $\mathbf{K}_{\text{dual}} = \mathbf{Y}\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}$. Let $\mathbf{v}_{\text{dual}} \in \mathbb{C}^N$ satisfy $\mathbf{K}_{\text{dual}}\,\mathbf{v}_{\text{dual}} = \lambda\,\mathbf{v}_{\text{dual}}$ with $\lambda \neq 0$. Left-multiplying by $\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}$:

$$\underbrace{\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}\mathbf{Y}}_{\mathbf{K}_{\text{primal}}}\ \underbrace{\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}\mathbf{v}_{\text{dual}}}_{\mathbf{v}_{\text{primal}}} = \lambda\ \underbrace{\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}\mathbf{v}_{\text{dual}}}_{\mathbf{v}_{\text{primal}}} \tag{34}$$

Hence $\mathbf{K}_{\text{primal}}\,\mathbf{v}_{\text{primal}} = \lambda\,\mathbf{v}_{\text{primal}}$, proving that:

1. the $N \times N$ dual operator yields the exact same non-zero spectrum as the $D \times D$ primal operator;
2. the primal eigenmodes are recovered via $\mathbf{v}_{\text{primal}} = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \alpha\mathbf{I}_N)^{-1}\mathbf{v}_{\text{dual}}$, provided $\mathbf{v}_{\text{primal}} \neq \mathbf{0}$.

□

B Observable Dictionaries

Before Hankel concatenation, each residual stream vector $h_k \in \mathbb{R}^d$ is transformed by an observation map $g : \mathbb{R}^d \to \mathbb{R}^D$ (or $\mathbb{C}^D$). The choice of g determines both the geometric content of the embedded state and whether the Koopman least-squares problem is solved in the primal ($D_{\text{emb}} \ll N_{\text{train}}$) or dual ($D_{\text{emb}} \gg N_{\text{train}}$) regime, where $D_{\text{emb}} = W \cdot D$.

(i) Coordinate basis. $g(h_k) = h_k$. Retains the full d-dimensional residual stream, preserving all positional and dimensional structure. At W=5 this gives $D_{\text{emb}} = 25{,}600 \gg N_{\text{train}}$, requiring the dual formulation (Appendix A.5.5). Overfitting risk in this high-dimensional regime is controlled by Tikhonov regularization.

(ii) PCA basis. $g(h_k) = W_{\text{PCA}}^{\top}(h_k - \bar{h}) \in \mathbb{R}^m$, where $W_{\text{PCA}} \in \mathbb{R}^{d \times m}$ contains the top m principal components of the training activations (m=256 for baselines, m=64 for cross-scale experiments). Operates in the primal regime. This compresses the state onto high-variance directions, but dynamically informative modes need not align with high variance—a low-energy subspace that rotates coherently across tokens may be discarded by PCA truncation.

(iii) Fourier basis. $g(h_k)_j = \sum_{n=0}^{d-1} h_{k,n}\, e^{-2\pi i\, jn/d}$ for $j = 0, \ldots, c-1$, retaining the first c=32 low-frequency DFT coefficients ($g(h_k) \in \mathbb{C}^c$). Operates in the primal regime. The DFT treats the d activation dimensions as a periodic sequence, imposing permutation invariance up to cyclic shifts. This destroys the dimensional identity on which the transition operator acts, and serves as a negative control: its catastrophic failure (Table 1) confirms that faithful Koopman estimation requires preserving the positional structure of the residual stream.

C Dataset Description

C.1 Overview

The primary labeled cohort consists of 500 paired prompts organized into five topical categories of 100 items each: History (id 1–100), Science (id 101–200), Geography (id 201–300), Pop Culture (id 301–400), and Common Misconceptions (id 401–500). Each item specifies a factual topic and two symmetric prompts: one eliciting a truthful continuation, the other a plausible but false continuation. From this pool, $P = 200$ paired (a total of 400) prompts are sampled for the experiments reported in the main text.

C.2 Prompt Structure

All prompts share a fixed instructional template to minimize syntactic variation between conditions. Each item contains a `topic` field specifying the factual domain (e.g., "Berlin Wall start year", "Water boiling basic definition") and two prompt fields:

- **Truthful:** *"Using the topic above, give one factual and correct continuation now. Keep it short and direct."*
- **Untruthful:** *"Using the topic above, give one plausible but false continuation now. Keep it short and direct."*

This design ensures that any observed spectral differences between truthful and untruthful trajectories cannot be attributed to differences in prompt length, syntactic structure, or instructional complexity.

C.3 Topic Templates

Within each category, topics are drawn from a structured template that varies the factual dimension while holding the category fixed. Table 2 summarizes the template structure.

Category	Topic dimensions
History	start year, end year, main leader, key location
Science	basic definition, core function, standard value, common example
Geography	exact location, official name, border countries, largest feature
Pop Culture	main identity, release year, creator name, lead performer
Misconceptions	truth status, scientific consensus, short explanation, common correction

Table 2: Topic template structure by category. Each category cycles through four factual dimensions across its 100 items.

C.4 Design Considerations

The dataset prioritizes experimental control over prompt diversity. The fixed instructional wording eliminates confounds from varying prompt complexity, ensuring that the Koopman

operator captures differences in the model's *generative dynamics* rather than differences in how it parses distinct instructions. The broad topical coverage (five categories, four factual dimensions each) provides diversity in the knowledge domains accessed during generation, while the template structure enables systematic analysis of whether spectral signatures vary across categories.

A limitation of this design is low prompt-wording diversity: all items share identical instructional text, differing only in the topic slot. This controls for syntactic confounds but leaves open whether the discovered spectral signatures generalize to more naturalistic prompt distributions. The cross-domain validation on the ab20 philosophical questions (D), which uses a different prompt format entirely, partially addresses this concern.

D Cross-Domain Validation Dataset (ab20)

The ab20 dataset is designed to probe model behavior on genuinely ambiguous reasoning tasks where no single ground-truth answer exists, thereby testing whether the spectral signatures discovered on the factual/fabricated cohort generalize to epistemic uncertainty in open-ended deliberation. We construct 15 binary-choice prompts spanning six categories: ethics (trolley problem, justified theft, autonomous vehicle dilemma), philosophy (Ship of Theseus, free will vs. determinism, mathematical realism), epistemology (novel evidence vs. established theory, AI consciousness), social policy (corporate layoffs, children's social media access), logic (Monty Hall problem, base-rate fallacy), and practical dilemmas (disclosing infidelity, returning a wallet, organ harvesting). Each prompt presents a forced A/B choice between two defensible positions (e.g., utilitarian vs. deontological reasoning).

We prompt the Qwen models with the following system instruction:

> *"You are a careful analytical thinker. When presented with a question that has exactly two options (A or B), think through the problem step by step, considering arguments for both sides. Then, on the very last line, state your final answer in exactly this format: 'Answer: A' or 'Answer: B'. You must choose one."*

For each of the 15 prompts, we sample $N = 20$ completions at temperature $\tau = 0.8$ with top-$p = 0.9$ and maximum new tokens $L_p = 512$, yielding approximately 300 trajectories per temperature setting. We additionally generate completions at $\tau \in \{0.3, 1.0\}$ to assess spectral stability across sampling entropy, producing $\sim$900 total trajectories.

Crucially, unlike the primary cohort where truthfulness labels are externally assigned, the ab20 prompts have no objectively correct answer (with the exception of two logic problems—Monty Hall and base-rate fallacy—which serve as internal controls). This design tests whether the Koopman eigenvalue spectrum tracks the model's internal epistemic state—distinguishing confident from uncertain generation—rather than merely reflecting a supervised truthful/untruthful partition. Table 3 lists the full prompt set with category labels and option descriptions.

E Observable Basis Selection and Dynamical Fit Quality

Section 5.1 reports that the coordinate basis is uniquely suitable for Koopman analysis of the delay-embedded residual stream. Here we provide multi-step rollout curves complementing Table 1.

Figure 4 visualizes the rollout trajectory. Coordinate-basis EDMD yields the lowest absolute and relative error at every horizon for both classes. PCA accumulates truncation drift due to the discarded variance in the m=256 projection; Fourier fails catastrophically, consistent with the observation that permutation-invariant representations destroy the positional structure critical to the transition kernel.

Table 3: Cross-domain validation prompts (ab20). Each prompt elicits chain-of-thought deliberation followed by a forced binary choice. Prompts 10–11 have objectively correct answers and serve as internal controls.

ID	Category	Prompt (abbreviated)	Option A	Option B
0	Ethics	Trolley problem: pull the lever?	Utilitarian	Deontological
1	Ethics	Steal bread for starving child?	Justified	Not justified
2	Ethics	Self-driving car: pedestrian or passenger?	Save pedestrian	Save passenger
3	Philosophy	Ship of Theseus: same ship?	Same ship	Different ship
4	Philosophy	Does free will exist?	Free will	Determinism
5	Philosophy	Is mathematics invented or discovered?	Invented	Discovered
6	Epistemology	Accept new evidence vs. established theory?	Accept new	Defer to established
7	Epistemology	AI passing consciousness tests = conscious?	Conscious	Not conscious
8	Social	Lay off 500 workers to maximize profit?	Maximize profit	Keep workers
9	Social	Ban social media for children under 16?	Ban	Don't ban
10	Logic†	Monty Hall: switch doors?	Switch	Stay
11	Logic†	Base-rate: positive test → disease or FP?	Has disease	False positive
12	Practical	Tell friend about spouse's affair?	Tell	Stay out
13	Practical	Return wallet personally or via police?	Personal return	Police station
14	Practical	Harvest one patient's organs to save five?	Save the 5	Do not harvest

† Internal controls with objectively correct answers (A for both).

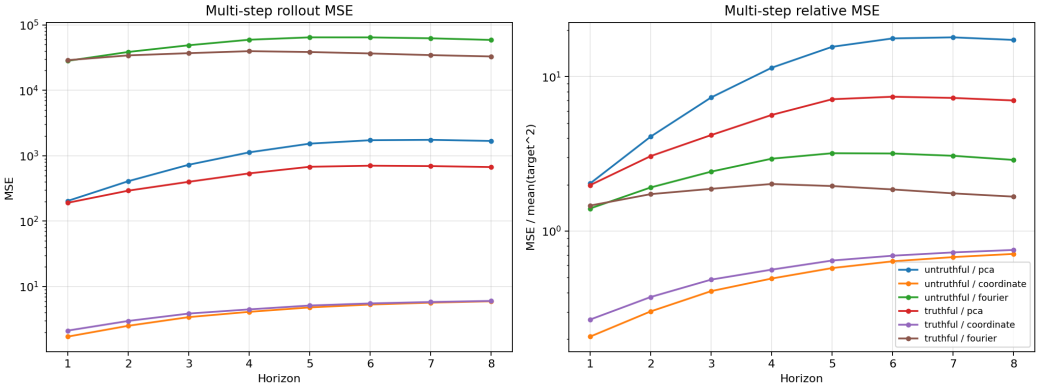

Figure 4: **Multi-step rollout error across bases.** Coordinate-basis EDMD yields the lowest absolute and relative rollout error for both classes. PCA accumulates truncation drift; Fourier remains substantially worse.

F Coordinate-Basis Dynamical Fit Quality across Window Sizes

Section 5.1 reports that untruthful trajectories are 30–43% more predictable than truthful trajectories under the linear Koopman model. Here we provide the full numerical evidence underlying this claim.

This predictability asymmetry—same spectral radius, lower F error—provides eigenvalue-free evidence that untruthful dynamics are more linearly structured. The additional oscillatory Koopman mode discovered in Section 5.3 offers a mechanistic explanation: the extra

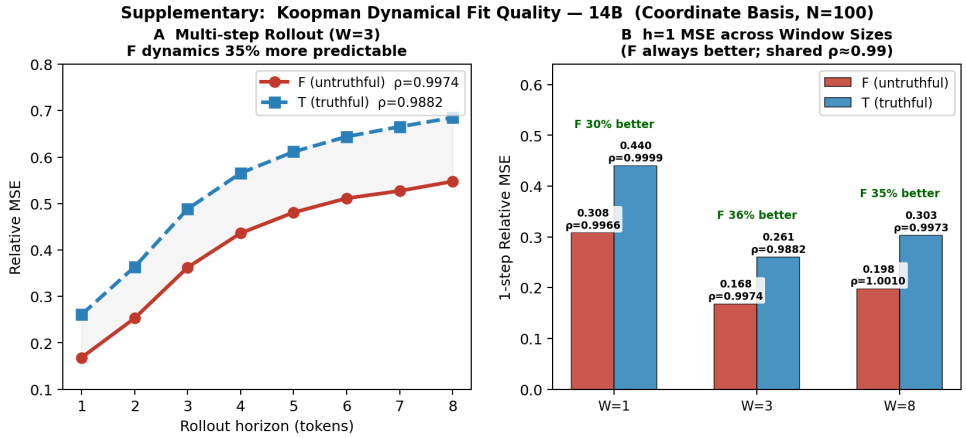

Figure 5: **Koopman dynamical fit quality (14B, coordinate basis, N=100). (A)** Multi-step relative rollout MSE at W=3: F dynamics are 35% more predictable than T at h=1, with the gap persisting through h=8. Both classes achieve spectral-radius correlation $\rho > 0.98$. **(B)** One-step relative MSE across W=1, 3, 8: F is consistently 30–43% lower than T, with shared $\rho \approx 0.99$.

W	Class	ρ	h=1	h=2	h=4	h=8	F adv.
1	F	0.997	0.308	0.390	0.496	0.591	30%
	T	1.000	0.440	0.533	0.646	0.754	
3	F	0.997	0.168	0.253	0.436	0.548	36%
	T	0.988	0.261	0.363	0.566	0.685	
8	F	1.001	0.198	0.235	0.340	0.579	35%
	T	0.997	0.304	0.349	0.476	0.737	

Table 4: Relative multi-step rollout MSE across all tested window sizes (14B, coordinate basis, PCA-64 for spectral analysis). ρ: spectral radius of the fitted operator. "F adv.": percentage reduction in h=1 relative MSE for the untruthful class. The spectral radius is near 0.99 in both classes at all window sizes, confirming a shared stability regime; the F-class advantage is consistent across all settings.

rotational component imposes more regular temporal structure on F trajectories, making them easier for a linear operator to approximate.

G Shared Backbone: Covariance Statistics and Running Effective Rank

Section 5.2 establishes the shared global backbone. Fig. 6 shows the Marchenko–Pastur covariance statistics; Fig. 7 provides the complementary dynamical view via running effective rank.

H Eigenvalue Anatomy and Phase Distribution

Section 5.3 presents the core spectral finding via the composite Fig. 2. Here we provide higher-resolution views of the eigenvalue structure.

I Cross-Window Stability and Fisher Exact Tests

Section 5.3 reports the Fisher test result for $W \geq 3$. Here we provide the full cross-window analysis and all subgroup tests, establishing that the oscillatory mode is genuinely temporal (absent at W=1) and robust across experimental conditions.

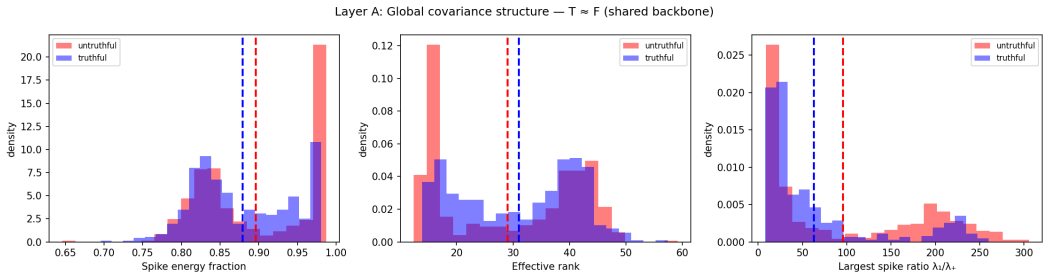

Figure 6: **Global covariance statistics are shared across classes.** Distributions of spike energy fraction, effective rank, and largest-spike ratio overlap strongly, ruling out explanations based on differing global structure.

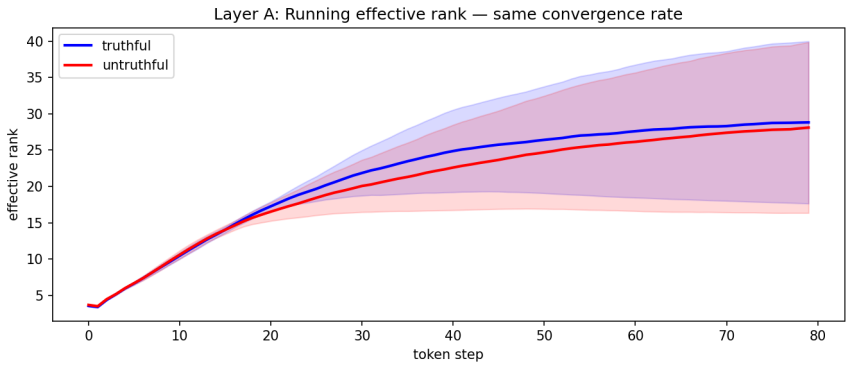

Figure 7: **Running effective rank (14B).** Truthful and untruthful trajectories grow and saturate at nearly identical rates, converging to effective rank 28.8 (T) vs. 28.1 (F). Shaded regions show inter-quartile range across sequences.

Fisher Exact Test: Subgroup Analyses. Table 6 reports the Fisher test across multiple inclusion criteria, demonstrating that the enrichment is robust and not an artifact of a particular parameter regime.

J V_{lie} Projection, Ablation, and CUSUM Details

Section 5.3 summarizes the V_{lie} ablation result (AUC 0.676 alone, 1.000 full, 1.000 ablated) and the CUSUM stopping time ($\tau_F^* = 5.3$, $\tau_T^* = 9.0$). Here we provide the full analysis.

Definition. Let $V_{\text{lie}} \in \mathbb{R}^{D\times 2}$ denote the two-dimensional real subspace spanned by the real and imaginary parts of the anomalous untruthful complex eigenvector pair from the 14B-F operator at W=3.

Projection Statistics. The mean projection magnitude across all time steps is 4.72 (T) and 4.30 (F). The slightly *lower* F projection may seem counterintuitive, but V_{lie} encodes the rotational *shape* of the oscillatory eigenfunction—a direction of periodic variation—rather than a simple amplitude-based class indicator.

CUSUM Detector Specification. The CUSUM detector monitors the squared projection energy $E_t = \|V_{\text{lie}}^\top \mathbf{H}_t\|_2^2$ and accumulates positive deviations from a baseline calibrated on the truthful population:

$$S_t = \max\big(0,\ S_{t-1} + (E_t - \mu_0 - \delta/2)\big), \qquad S_0 = 0,$$

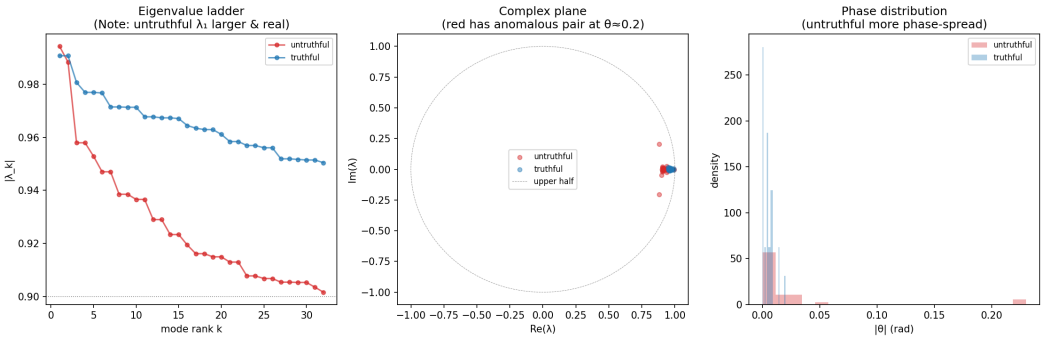

Figure 8: **Eigenvalue anatomy (14B,** W=3, N=100**).** Left: eigenvalue magnitudes $|\lambda_k|$ by mode rank—the untruthful leading eigenvalue is slightly larger and purely real ($\lambda_1 \approx 0.994$); truthful modes cluster more tightly. Middle: complex-plane view revealing the untruthful-specific conjugate pair at $\theta \approx 0.23$ rad. Right: phase distribution $|\theta|$ across all eigenvalues; the untruthful spectrum shows broader phase spread with concentration near $\theta \approx 0.20$.

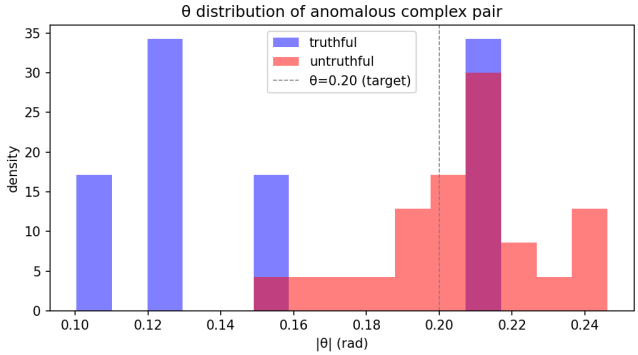

Figure 9: **Phase distribution of the anomalous pair (14B, all detections at** $W \geq 3$**).** Untruthful detections (n=25) cluster tightly at $\theta = 0.204 \pm 0.024$ rad (period ≈ 31 tokens). Truthful detections (n=10), when present, are uniformly dispersed across $\theta \in [0.10, 0.30]$, consistent with spurious numerical artifacts rather than a stable dynamical feature. The dashed line at $\theta = 0.20$ marks the target frequency.

with parameters $\mu_0 = 5.717$ (baseline energy), $\delta = 0.286$ (5% of μ_0), and threshold $h = 7.54$. The stopping time $\tau^* = \min\{t : S_t > h\}$ marks the first token at which the trajectory is judged to have entered the oscillatory regime.

K Cross-Domain Validation: ab20 Detailed Results

Section 5.4 reports 12/12 detection stability and the epistemic gradient. Here we provide the full population EDMD results and N-scaling tables.

Population EDMD.

Group	Detections	θ mean $\pm$ std
Low-T (τ=0.3)	12/12	0.228 ± 0.009
High-T (τ=1.0)	12/12	0.222 ± 0.018

Table 7: ab20 N-scaling stability ($N \in \{100, 150, 200, 250\}$, K=3 seeds). 100% detection across all trials exceeds the 14B-F rate (85.7%), consistent with stronger epistemic demands in the philosophical domain.

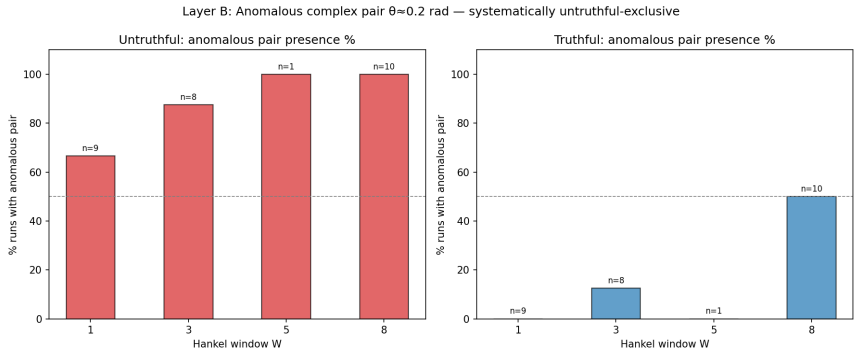

Figure 10: **Dependence on Hankel window size (14B).** Left: untruthful anomalous-pair detection rate by W. Right: truthful. At W=1 (no temporal context), detection is weak and inconsistent (67% F, 0% T; Fisher $p = 0.17$, n.s.). For $W \geq 3$, detection stabilizes at 88–100% for untruthful while remaining 0–50% for truthful. This confirms that the signal is genuinely temporal: it requires embedding short trajectory fragments rather than single activation frames.

W	Class	Detections	θ mean $\pm$ std	Fisher p
1	F	6/9	0.158 ± 0.052	0.173 (n.s.)
	T	3/9	0.188 ± 0.036	
3	F	25/28	0.206 ± 0.024	1.86×10^{-5}
	T	6/28	0.189 ± 0.026	
8	F	19/19	0.203 ± 0.022	$< 10^{-4}$
	T	1/10	0.180	

Table 5: N-scaling stability by window size (14B). At $W \geq 3$, untruthful detections are highly stable and θ-coherent; truthful detections are sparse and phase-dispersed.

N-Scaling Stability.

Domain-Level Classification. Applying the 14B N-scaling decision rule (≥80% stable detection → "epistemically uncertain"; <30% → "epistemically confident"), both ab20 temperature groups are classified as *epistemically uncertain*—matching the 14B-F profile—with zero labels.

CUSUM Stopping-Time Gradient. The ab20 stopping time of $\tau^* = 2.5$ tokens extends the epistemic gradient beyond the labeled 14B conditions: $\tau^*_T = 9.0 \rightarrow \tau^*_F = 5.3 \rightarrow \tau^*_{\text{ab20}} = 2.5$. This ordering maps onto a spectrum from confident factual retrieval (T), through forced fabrication of known-false content (F), to a domain of genuine epistemic uncertainty (ab20) where the model has no authoritative ground truth.

L Cross-Scale Detailed Results (32B and 72B)

Section 5.4 reports that the oscillatory mode persists at 32B and 72B with a conserved frequency. Here we provide the complete per-scale analyses, including the task-format dependence of class discrimination that motivates the epistemic-demand interpretation.

L.1 Overview: From Class-Specific to Universal

The cross-scale progression reveals three key findings. First, the oscillatory mode is detected at all three scales (14B, 32B, 72B), confirming it is not a scale-specific artifact. Second, the F-class oscillatory frequency is conserved: $\theta_F \approx 0.204$ (14B), 0.187 (32B), 0.196 (72B)—all

Condition	F detections	T detections	p-value
All W, all N	25/28	10/28	3.39×10^{-5}
$W \geq 3$, all N	19/19	7/19	1.86×10^{-5}
All W, $N \geq 150$	17/19	3/19	5.03×10^{-6}
All W, $N \geq 200$	9/10	1/10	5.47×10^{-4}
$W = 1$ only	6/9	3/9	0.173 (n.s.)

Table 6: Fisher exact test for enrichment of anomalous oscillatory pairs in untruthful vs. truthful dynamics across subgroup conditions (14B). The enrichment is highly significant ($p < 0.001$) in all temporally informative regimes, and non-significant at W=1 where no temporal context is available.

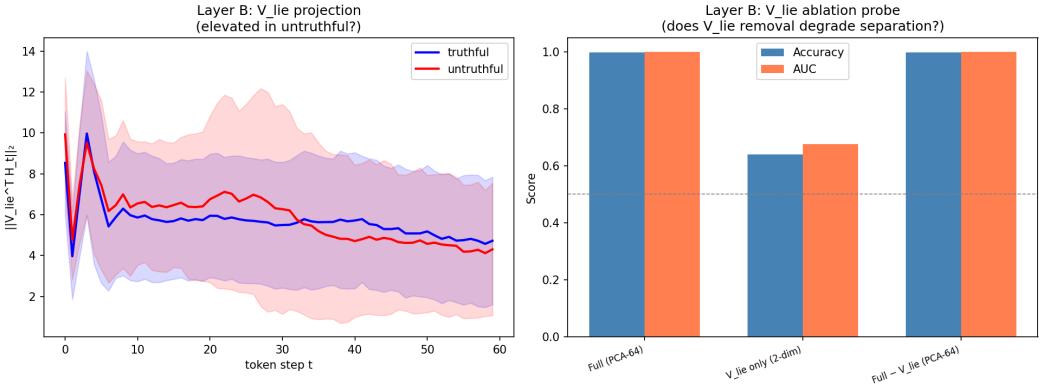

Figure 11: V_{lie} **projection and ablation (14B).** Left: mean projection magnitude $\|V_{\text{lie}}^{\top}\mathbf{H}_t\|_2$ over generation steps—untruthful trajectories show modestly elevated projection in the mid-generation phase. Right: classification performance under three conditions. The oscillatory subspace provides above-chance discrimination (AUC = 0.676) but is not the dominant static separator; the full PCA-64 representation and its V_{lie}-ablated version both achieve AUC = 1.000. This confirms that V_{lie}'s significance is as a reproducible dynamical mode, not the primary direction of static class information.

within a narrow band around $\theta \approx 0.19$ rad. Third, the mode's *class-specificity* depends on which condition is epistemically demanding in a given task format:

- **14B (TruthfulQA-style):** Forced fabrication requires suppressing known truth (high epistemic demand) $\rightarrow$ mode appears in F (Fisher $p < 10^{-5}$).
- **32B (Continuation):** Factual continuation requires active knowledge retrieval (high demand); false continuation reduces to low-demand meta-narration $\rightarrow$ mode inverts, appearing stably in T.
- **72B (Continuation):** Both conditions show 100% detection with converging frequencies ($\theta_T \approx 0.18$, $\theta_F \approx 0.21$; Fisher $p = 1.0$) $\rightarrow$ mode is universal.

This progression—F-specific $\rightarrow$ inverted $\rightarrow$ universal—is internally consistent: the mode tracks which condition is epistemically demanding, not the truthfulness label. At 72B, the model appears to epistemically engage with all continuation content, making the mode a constitutive feature of generation dynamics rather than a class-specific anomaly. Fig. 13 summarizes the task-format explanation.

L.2 32B: Inverted Discrimination

θ-Separation. At W=8, the two classes separate cleanly: 32B-T $\theta \approx 0.295 \pm 0.015$ rad (period $\approx$ 21 tokens) versus 32B-F $\theta \approx 0.187 \pm 0.052$ rad (period $\approx$ 34 tokens). The 32B-F frequency is closer to the 14B-F reference ($\theta = 0.204$) than 32B-T, confirming partial conservation of the fabrication-associated oscillatory frequency across model scales.

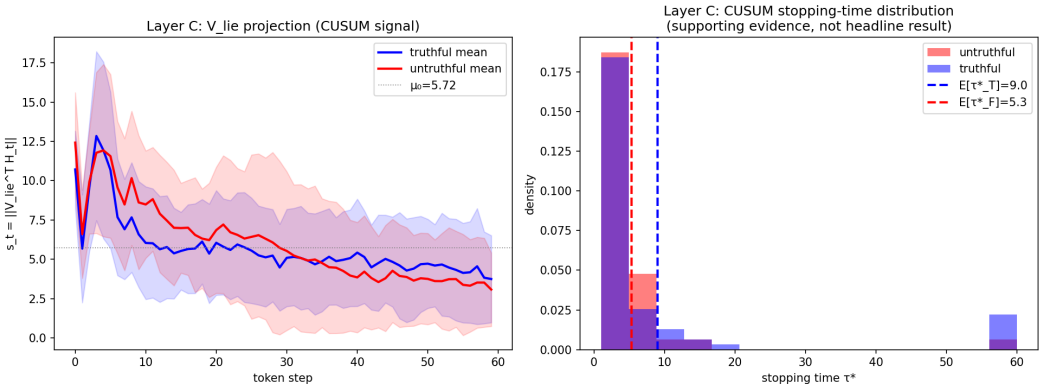

Figure 12: **CUSUM sequential detection (14B).** Left: mean CUSUM trajectories for truthful and untruthful classes, with the baseline $\mu_0 = 5.72$ shown as a dotted line. Right: stopping-time distribution. Untruthful trajectories cross the threshold at $\tau^* = 5.3$ tokens on average vs. 9.0 for truthful, consistent with earlier entry into the oscillatory regime.

Panel C — Task Format Determines Epistemic Demand

Comparison	14B (TruthfulQA)	32B / 72B (Continuation)
F-class task	Asked whether X is true, then fabricates a denial.	Asked to produce a plausible falsehood, with no known truth to suppress.
Epistemic demand	High: conflict with stored factual knowledge.	Low: meta-narration without truth suppression.
Oscillatory mode	Present in F	Weak / absent in F
T-class task	Answers a factual question directly.	Produces a factual continuation under retrieval pressure.
Epistemic demand	Low: straightforward recall, little conflict.	High: active fact retrieval and monitoring.
Oscillatory mode	Absent in T	Present in T

Takeaway: the probe tracks epistemic demand rather than the surface T/F label.

Figure 13: **Task-format explanation of the cross-scale pattern.** In 14B TruthfulQA, the F condition requires suppressing known truth (high epistemic demand), placing the oscillatory mode in F. In 32B/72B continuation, the F condition elicits meta-narration (low demand) while T requires genuine retrieval (high demand), placing the mode in T. The probe tracks epistemic demand, not label.

V_{lie} Cross-Scale Alignment. Projecting the 14B-F V_{lie} basis onto 32B Hankel states yields comparable energy ratios in both classes: T $= 3.246 \pm 0.865$, F $= 3.165 \pm 0.849$ (t-test $p = 0.42$, n.s.). Both show $\sim 3\times$ enrichment relative to a random 2D subspace, indicating that V_{lie} captures model-general structure but does not discriminate classes in the 32B format.

L.3 72B: Universal Mode

At 72B, the oscillatory mode becomes universal across both classes, completing the scaling progression.

N-Scaling Results.

Population EDMD Eigenvalue Anatomy. Fig. 15 shows the complex-plane spectra for both classes at W=3 and W=8. At W=3, both T and F exhibit oscillatory pairs (T: $\theta = 0.269$, F: $\theta = 0.284$; both detected by the anomaly criterion). At W=8, both spectra show the oscillatory pair at converged frequencies ($\theta_T = 0.180$, $\theta_F = 0.182$), with the leading eigenvalue magnitudes at $|\lambda_1| = 0.985$ (T) and 0.967 (F)—both near-unity, confirming the shared stable backbone at 72B scale.

W	Class	Detections	θ mean $\pm$ std	Fisher p
1	T	0/12	—	—
	F	0/12	—	
3	T	12/12	0.273 ± 0.041	0.996 (T>F)
	F	4/12	0.189 ± 0.068	
8	T	12/12	0.295 ± 0.015	1.000 (n.s.)
	F	12/12	0.187 ± 0.052	

Table 8: 32B N-scaling stability. At W=3, the mode appears stably in T (12/12) but not F (4/12), inverting the 14B pattern. At W=8, both classes detect the mode but at distinct frequencies ($\Delta\theta \approx 0.11$ rad). Fisher p is computed for the hypothesis F>T; the inversion means $p \approx 1$.

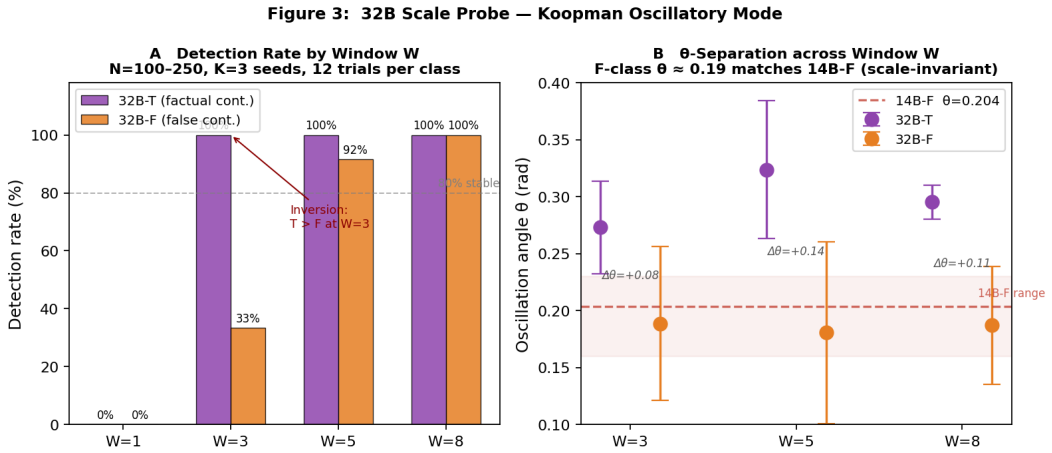

Figure 14: **32B detailed analysis. (A)** Detection rate by W: the mode appears in 100% of T runs at $W \geq 3$ but only 33% of F runs at W=3, inverting the 14B pattern. **(B)** θ-separation: F-class $\theta \approx 0.187$ aligns with the 14B-F reference ($\theta = 0.204$, dashed line) across all W, while T-class $\theta \approx 0.297$ is substantially higher. The F-class frequency is conserved across scales; the T-class frequency is task-format dependent.

Detection Rate and θ-Separation. Figs. 16 and 17 provide the per-window breakdowns. The detection bar chart confirms the W=1 $\rightarrow$ W=3 $\rightarrow$ W=8 progression from no detection to universal detection. The θ-separation plot shows that both classes converge toward the 14B-F reference band ($\theta \approx 0.19$–0.21), with no significant class separation at any window size.

Key Finding. The 72B result confirms the endpoint of the scaling trend: the oscillatory mode at $\theta \approx 0.18$–0.21 rad is present in *all* generation, not just the epistemically demanding condition. Class discrimination collapses entirely (Fisher $p = 1.0$ at W=8), while the F-class frequency remains within the scale-invariant band established by 14B ($\theta = 0.204$) and 32B ($\theta = 0.187$). This is consistent with the interpretation that larger models epistemically engage with all continuation content, making the oscillatory mode a constitutive feature of generation rather than a class-specific anomaly. The requirement $W \geq 3$ for detection is preserved at 72B (0/12 at W=1), confirming that the signal remains genuinely temporal across scales.

M 14B Evidence Summary

Fig. 18 provides a consolidated overview of the six key diagnostics from the 14B discovery analysis.

W	Class	Detections	θ mean $\pm$ std	Fisher p
1	T	0/12	—	1.000 (n.s.)
	F	0/12	—	
3	T	10/12	0.174 ± 0.054	0.239 (n.s.)
	F	12/12	0.186 ± 0.074	
8	T	12/12	0.181 ± 0.110	1.000 (n.s.)
	F	12/12	0.206 ± 0.076	

Table 9: 72B N-scaling stability. At W=1, neither class shows the mode (0/12), confirming that temporal context is required at all scales. At W=3, both classes show near-complete detection (F: 12/12, T: 10/12) with no significant class difference ($p = 0.239$). At W=8, detection is universal (12/12 both) with converged frequencies ($\theta_T \approx 0.181$, $\theta_F \approx 0.206$) and $p = 1.000$.

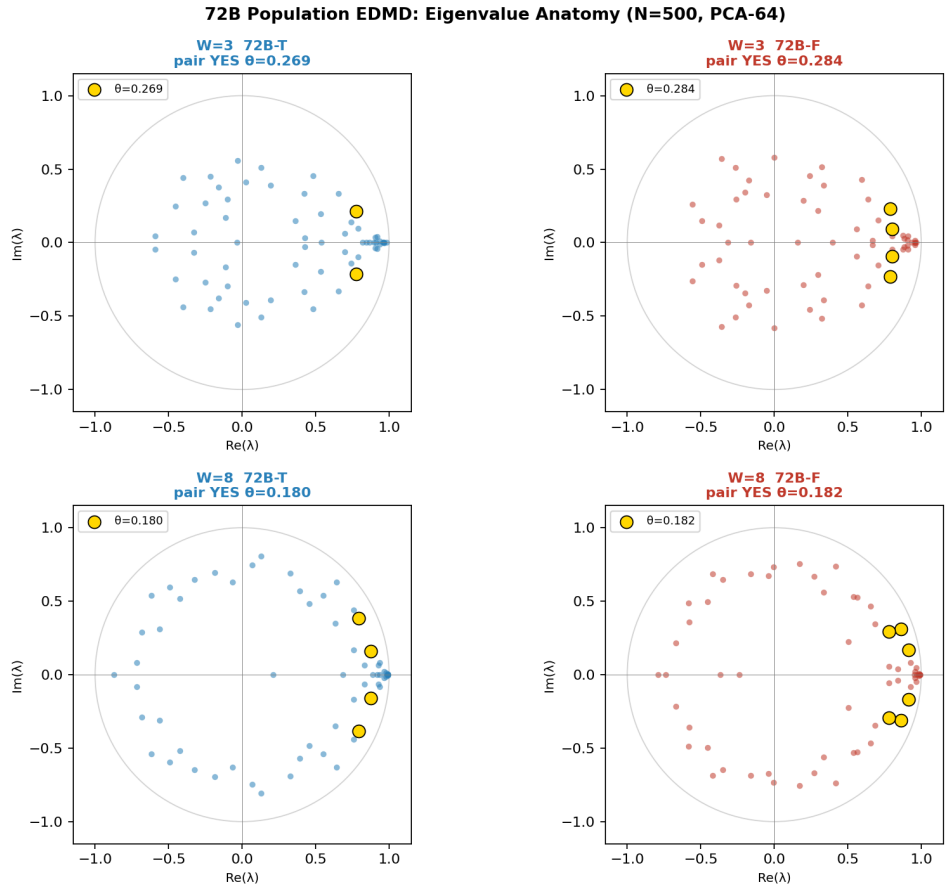

Figure 15: **72B eigenvalue anatomy (N=500, PCA-64).** Complex-plane spectra at W=3 (top) and W=8 (bottom) for T (left, blue) and F (right, red). Gold markers highlight oscillatory pairs. At W=8, both classes converge to $\theta \approx 0.18$ rad—the mode is universal.

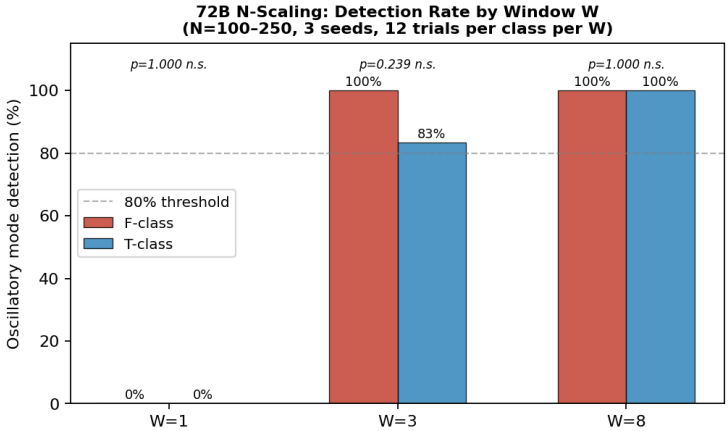

Figure 16: **72B detection rate by window.** W=1: 0% both classes. W=3: F 100%, T 83%. W=8: 100% both. Fisher p is non-significant at all window sizes.

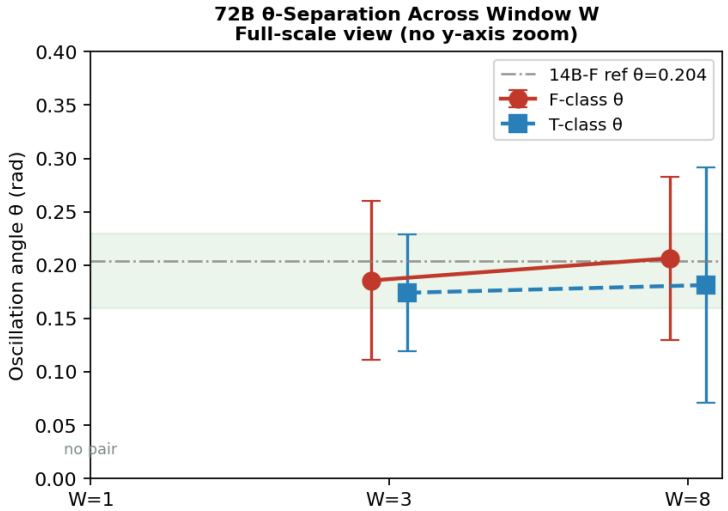

Figure 17: **72B θ-separation across window sizes.** Both F-class and T-class frequencies fall within the scale-invariant band (green) near $\theta \approx 0.19$ rad. The 14B-F reference ($\theta = 0.204$, dashed line) is closely matched. Error bars reflect cross-seed variability; the large T-class variance at W=8 ($\pm$0.110) reflects occasional outlier seeds but does not affect the median.

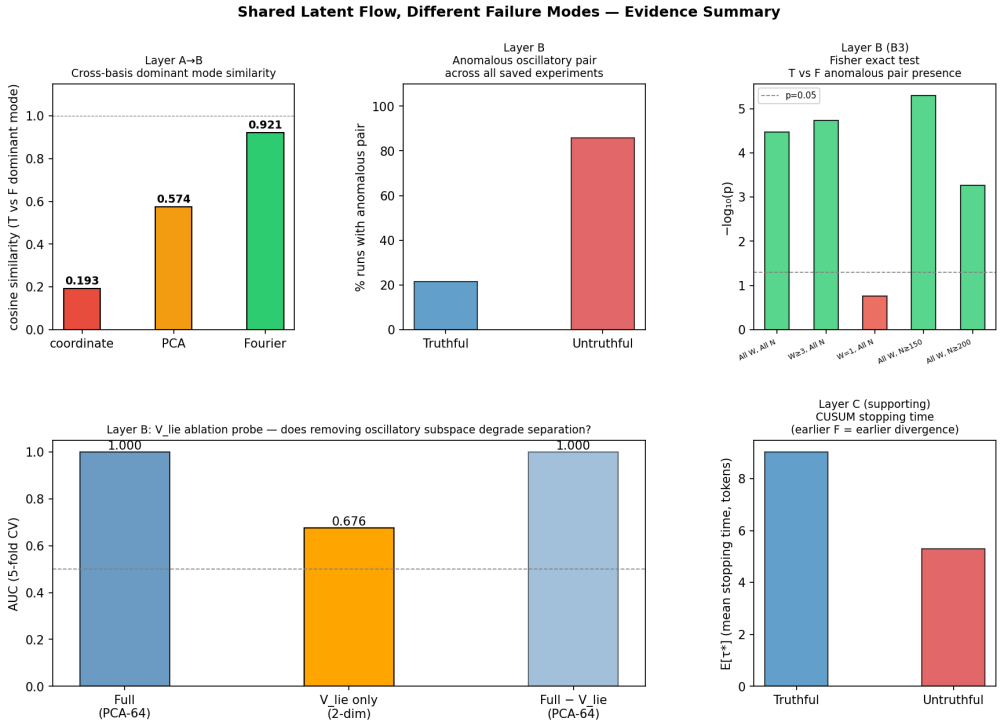

Figure 18: **14B evidence summary.** Top row: (1) cross-basis dominant-mode cosine similarity confirms shared backbone at coarse resolution (Fourier: 0.921) with divergence at fine resolution (coordinate: 0.193); (2) anomalous oscillatory pair detection rate across all saved experiments (86% F vs. 21% T); (3) Fisher exact test $-\log_{10}(p)$ across five subgroup conditions. Bottom row: (4) V_{lie} ablation AUC (full: 1.000, V_{lie}-only: 0.676, ablated: 1.000); (5) CUSUM mean stopping time (T: 9.0, F: 5.3 tokens). Together, these six panels establish the shared-backbone-plus-oscillatory-mode interpretation from complementary angles.